



Function and Disorders of the Parathyroid Gland

Note: This reference document contains the content of the Canvas page for “Function and Disorders of the Parathyroid Gland” with minor modifications for presentation in a document format. Refer to this session’s Canvas page for the full interactive experience or if any hyperlinks do not function properly.

Table of Contents

Table of Contents	1
Overview	4
Learning Objectives	4
Preparation Content	5
Function and Disorders of the Parathyroid Gland Part 1 of 4: Structure and Function of the Parathyroid Glands.....	6
Overview	6
Learning Objectives	6
Gross Anatomy/ Microanatomy	6
The Parathyroid Glands	7
Location	7
Vascular Supply.....	8
Venous Drainage	8
Development of the Parathyroid Glands.....	9
Developmental Anomalies	9
Structure of the Parathyroid Glands.....	10
Physiology and Pathology.....	11
How Is Parathyroid Hormone Synthesized?	11
How Does Parathyroid Hormone Act on Tissues?	12
How Is Parathyroid Hormone Secretion Regulated?	14
Summary	16
Check Your Knowledge	17
Part 1 Structure and Function of the Parathyroid Glands Review Questions.....	17
Part 1 Structure and Function of the Parathyroid Glands Review Questions and Answers	18
Function and Disorders of the Parathyroid Gland Part 2 of 4: Hypocalcemia and Hypercalcemia ...	20



Overview	20
Learning Objectives	20
Hypocalcemia	21
What Is Hypocalcemia?	21
How Do Patients With Hypocalcemia Present?	21
How do we elicit these signs?	22
What is the Pathophysiology of Hypocalcemia?	23
How Do We Diagnose Hypocalcemia?	27
How Do We Manage Hypocalcemia?	28
Summary of Hypocalcemia	29
Hypercalcemia	30
What Is Hypercalcemia?	30
How Do Patients With Hypercalcemia Present?	30
What Is the Pathophysiology of Hypercalcemia?	32
How Do We Diagnose Hypercalcemia?	36
How Do We Manage Hypercalcemia?	37
Summary of Hypercalcemia	40
Check Your Knowledge	40
Part 2 Hypocalcemia and Hypercalcemia Review Questions	40
Part 2 Hypocalcemia and Hypercalcemia Review Questions and Answers	42
Function and Disorders of the Parathyroid Gland Part 3 of 4: Hypophosphatemia and Hyperphosphatemia	47
Overview	47
Learning Objectives	47
Hypophosphatemia	47
What Is Hypophosphatemia?	47
How Do Patients With Hypophosphatemia Present?	48
What Is the Pathophysiology of Hypophosphatemia?	49
How Do We Diagnose Hypophosphatemia?	53
How Do We Manage Hypophosphatemia?	54
Summary of Hypophosphatemia	54
Hyperphosphatemia	55
What Is Hyperphosphatemia?	55



How Do Patients With Hyperphosphatemia Present?	56
What Is the Pathophysiology of Hyperphosphatemia?	57
How Do We Diagnose Hyperphosphatemia?	59
How Do We Manage Hyperphosphatemia?	60
Summary of Hyperphosphatemia.....	61
Check Your Knowledge	62
Part 3 Hypophosphatemia and Hyperphosphatemia Review Questions	62
Part 3 Hypophosphatemia and Hyperphosphatemia Questions and Answers.....	64
Function and Disorders of the Parathyroid Gland Part 4 of 4: Hypoparathyroidism and Hyperparathyroidism	69
Overview	69
Learning Objectives	69
Hypoparathyroidism	70
What Is Hypoparathyroidism?	70
How Do Patients With Hypoparathyroidism Present?	70
What Causes Hypoparathyroidism?	71
How Do We Diagnose Hypoparathyroidism?	72
How Do We Manage Hypoparathyroidism?	73
Summary of Hypoparathyroidism	74
Hyperparathyroidism	75
Introduction.....	75
What Is Hyperparathyroidism?	75
How Do Patients With Hyperparathyroidism Present?	77
What Causes Hyperparathyroidism?	79
Hypercalcemia of Malignancy	80
How Do We Diagnose Hyperparathyroidism?	80
How Do We Manage Hyperparathyroidism?	82
Summary of Hyperparathyroidism.....	83
Check Your Knowledge	85
Part 4 Hypoparathyroidism and Hyperparathyroidism Review Questions	85
Part 4 Hypoparathyroidism and Hyperparathyroidism Review Questions and Answers	87



Overview

In this session, the normal and abnormal function of the parathyroid glands will be covered with emphasis on the role of the glands in maintaining normal calcium and phosphate levels and how these are altered in disease states. The diagnosis and clinical management of parathyroid related disorders will also be discussed.

Learning Objectives

By the end of this session, you will be able to meet the following learning objectives:

Gross & Microscopic Anatomy

1. Describe the gross anatomical features, vascular supply and drainage of the parathyroid glands.
2. Describe the development of the parathyroid glands.
3. Describe the histologic appearance of the parathyroid glands.

Physiology

1. Describe parathyroid hormone (PTH) synthesis, transport, and binding.
2. Identify the major physiological actions of parathyroid hormone.
3. Describe the regulation of PTH release.
4. Define hypocalcemia and the normal calcium reference range.
5. Define hypercalcemia and the normal calcium reference range.
6. Describe the physiologic effects of hypocalcemia on nerve and muscle cells and cardiac conduction.
7. Describe the physiologic effects of hypercalcemia on nerves, muscle cells, and cardiac conduction.

Pathology

1. Discuss the etiopathogenesis of hyperparathyroidism.
2. Explain the pathogenesis of brown tumor in the setting of chronic renal failure.
3. Understand the pathogenesis of 'hypercalcemia in malignancy. Differentiate the role of PTHrP from PTH in malignancy.
4. Describe the morphological features noted in i) Parathyroid adenoma, ii) Brown tumor. Differentiate parathyroid hyperplasia and carcinoma.

Clinical medicine

1. Discuss the clinical presentations of hyperparathyroidism and hypoparathyroidism.
2. Discuss the differential diagnoses for hypercalcemia and hypocalcemia.
3. Compare and contrast different clinical treatments for hypercalcemia.
4. Discuss investigative studies that are required for further evaluation of both hyperparathyroidism and hypoparathyroidism.



Pharmacology

1. Explain the pharmacological treatment options for hypocalcemia, including the mechanisms of action of calcium supplements and vitamin D analogs.
2. Evaluate the appropriate use of intravenous calcium in managing severe hypocalcemia.
3. Discuss the pharmacological management of hypercalcemia, including the roles of bisphosphonates, calcitonin, furosemide, and gallium nitrate.
4. Detail the pharmacological interventions for managing hypophosphatemia, including the use of oral and intravenous phosphate preparations and FGF23 antibody Burosumab.
5. Describe the role of phosphate binders in managing serum phosphate levels with a focus on mechanism of action and adverse effects.
6. Explain the use of PTH analogs and calcium/vitamin D supplementation in the management of hypoparathyroidism.
7. Discuss the pharmacological management of hyperparathyroidism, including the roles of bisphosphonates, cinacalcet and denosumab.

Preparation Content

For this session, content has been sectioned into separate pages. *Refer to the Canvas page for the full interactive experience of these sessions and quizzes.*

- Fusion Session | Workshop: Function and Disorders of the Parathyroid Gland **Part 1 of 4: Structure and Function of the Parathyroid Glands**
- Fusion Session | Workshop: Function and Disorders of the Parathyroid Gland **Part 2 of 4: Hypocalcemia and Hypercalcemia**
- Fusion Session | Workshop: Function and Disorders of the Parathyroid Gland **Part 3 of 4: Hypophosphatemia and Hyperphosphatemia**
- Fusion Session | Workshop: Function and Disorders of the Parathyroid Gland **Part 4 of 4: Hypoparathyroidism and Hyperparathyroidism**

Function and Disorders of the Parathyroid Gland Part 1 of 4: Structure and Function of the Parathyroid Glands

Overview

Unless otherwise noted, all content and images for this lecture are adapted from ScholarRx and used with permission.

Learning Objectives

This page covers the following learning objectives:

Gross & Microscopic Anatomy

1. Describe the gross anatomical features, vascular supply and drainage of the parathyroid glands.
2. Describe the development of the parathyroid glands.
3. Describe the histologic appearance of the parathyroid glands.

Physiology

1. Describe parathyroid hormone (PTH) synthesis, transport, and binding.
2. Identify the major physiological actions of parathyroid hormone.
3. Describe the regulation of PTH release.

Gross Anatomy/ Microanatomy

Case Connection

“The best way to describe it is like this: think of that video game where that little guy runs around eating up the little dots. Now think of a that little guy as a special cell that gets its instructions from parathyroid hormone. It obediently follows orders, goes to the bones, and eats them up. When it does that, calcium levels in the blood rise.” And so you begin your conversation with SF, a 43-year-old female who is about to undergo resection of parathyroid adenomas to treat her hypercalcemia. “So, now we remove the glands, called the parathyroid glands, which produce the hormone that sends that little guy to go to your bones.” “Wow, that’s really fascinating. Is that the only place in my body where this hormone works?” SF asks.

How would you respond to SF’s question? Consider your answer as you read, and we’ll revisit SF at the end of the brick.



The Parathyroid Glands

The parathyroid glands are part of your body's endocrine system, critical for maintaining bone health and serum calcium concentrations. They respond to low blood calcium levels by making and secreting parathyroid hormone, which regulates calcium homeostasis by releasing calcium from bone and increase bone turnover.

Location

Most individuals have four parathyroid glands (two superior and two inferior), but occasionally there are more in tissues of the neck or chest.

They are located between the true and false capsules of the thyroid gland on the posterior surface of the thyroid (a thick connective tissue capsule separates the glands from the thyroid tissue) (Figure 1).

- The false capsule is pretracheal fascia, which covers the trachea and also “encapsulates”—hence the term false capsule—the thyroid gland.
- The true capsule covers the thyroid deep to the false capsule and extends into the thyroid tissue via septae, creating delineations between thyroid lobes and lobules.

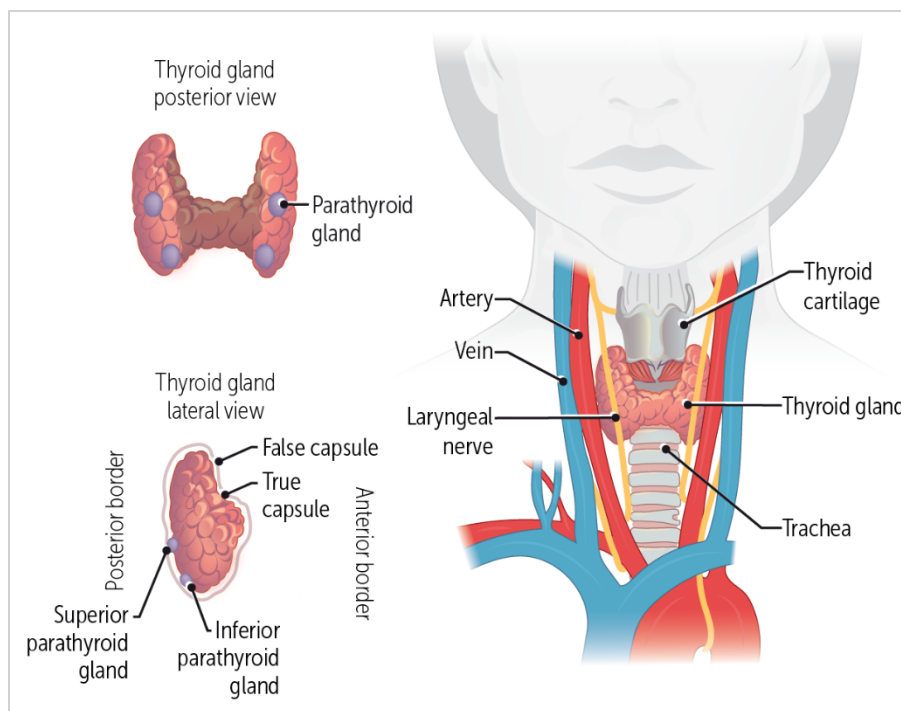


Figure 1 Image credit: ScholarRx

In healthy patients, the parathyroid glands are often as small as a grain of rice. Their specific locations varies from person to person. It is also possible to have more than four parathyroid glands. These extra glands can be found near the thyroid or be associated with the thymus.

Knowledge Check Question

Question: What hormone that regulates calcium homeostasis is secreted by the parathyroid glands?

Answer: Parathyroid hormone (PTH) regulates calcium homeostasis and is secreted by the parathyroid glands.

Vascular Supply

Because the parathyroid glands essentially adhere to the thyroid, they share a common blood supply. The inferior thyroid artery supplies blood to the parathyroid glands (Figure 2).

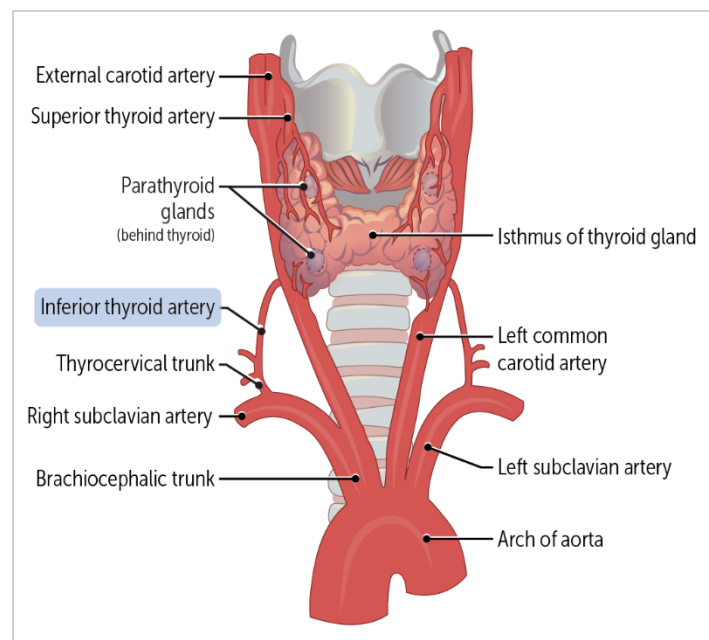


Figure 2 Image credit: ScholarRx

Venous Drainage

The venous drainage of the parathyroids is similar to that of the thyroid gland. Both the thyroid and the parathyroid glands are drained by the superior, middle, and inferior thyroid veins (Figure 3).

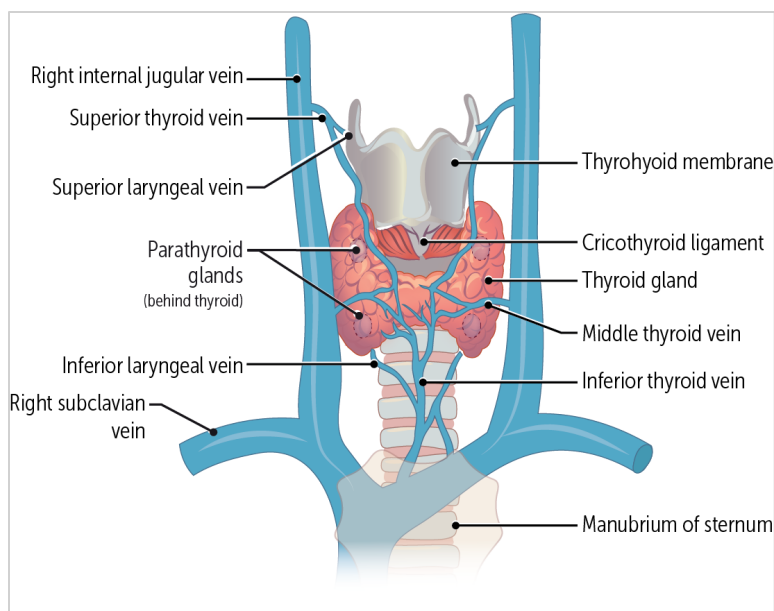


Figure 3 Image credit: ScholarRx

Knowledge Check Question

Question: What supplies blood to the parathyroid glands?

Answer: The inferior thyroid artery supplies blood to the parathyroid glands.

Development of the Parathyroid Glands

The parathyroid glands arise from the endoderm of the third and fourth pharyngeal pouches. The third pharyngeal pouch gives rise to the inferior parathyroids, and the fourth pharyngeal pouch gives rise to the superior parathyroids.

Developmental Anomalies

Because of the process of their descent during development, the parathyroid glands are subject to several abnormalities. Fragmentation of the glands during descent may give rise to accessory or supernumerary parathyroid glands, whereas an abnormal migration may lead to ectopic placement of the gland (e.g., inferior parathyroid glands in the anterior mediastinum with the thymus). Failure of the primordial cells to differentiate may lead to the absence of the parathyroid glands.

DiGeorge syndrome (also referred to as 22q11.2 deletion syndrome) occurs by a similar process and leads to congenital thymic aplasia and absent parathyroid glands. Patients have hypocalcemia from hypoparathyroidism, and the absent thymus leads to frequent infections resulting from a lack of T cells.



Structure of the Parathyroid Glands

The parathyroid glands are four small, round organs located on the posterior surface of the thyroid gland. They are composed of chief cells and oxyphil cells with some interspersed adipose cells (Figure 4).

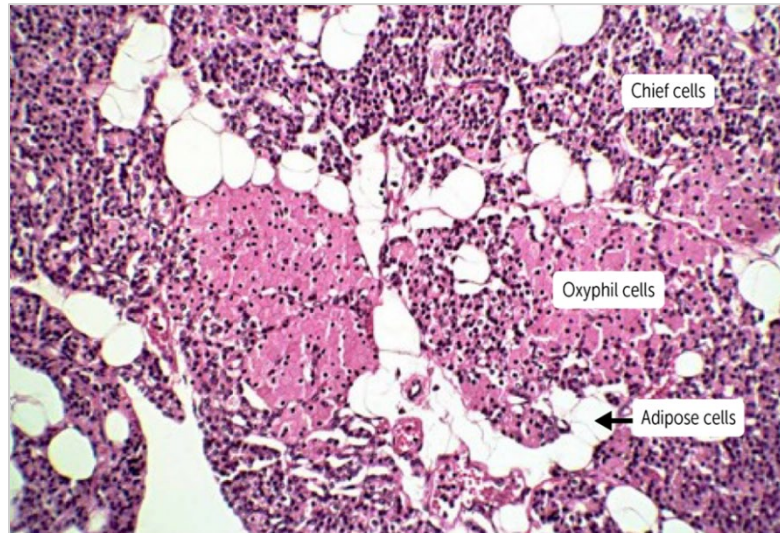


Figure 4 Credit: ©ScholarRx - Courtesy of Dr. Stephen Downing

Chief cells are compact cells with minimal cytoplasm that occur in sheets and vague groups. Although they look insignificant, they are what carry out the function of the parathyroid gland. They secrete a hormone called—not surprisingly—parathyroid hormone, which helps regulate serum calcium levels.

Oxyphil cells are large cells with abundant, eosinophilic cytoplasm that occur in clusters. They look like they should do something important, but their function is unknown.

Knowledge Check Question

Question: What types of cells are parathyroid glands composed of?

Answer: Parathyroid glands are composed of chief, oxyphil, and adipose cells.



Physiology and Pathology

How Is Parathyroid Hormone Synthesized?

Parathyroid hormone (PTH), a key regulator of calcium and phosphate homeostasis, is a peptide hormone secreted by the chief cells of the parathyroid gland. It is secreted as a preprohormone (110 amino acids) but is immediately cleaved into first the prohormone (90 amino acids), then the mature hormone (84 amino acids).

Transport and Receptor Binding

After synthesis in the parathyroid gland, PTH travels unbound in circulation. It binds to target tissues by using the PTH receptors (termed PTH1R and PTH2R). These are G protein-coupled receptors. They signal by using multiple pathways, including cAMP, phospholipase C, protein kinase C, and the release of intracellular calcium stores to alter gene expression.

Receptor Sites

As you'd expect from the functions of PTH, its receptors are found in high density in bones and cartilage. They are also found in renal tubules, where they control renal calcium and phosphate reabsorption.

Clinical Correlation



PTH-related peptides (PTHrP) secreted by some cancers can bind to the PTH receptors and mimic their effects, causing hypercalcemia and bone pain. This is an example of paraneoplastic syndrome, a metabolic effect caused by a substance secreted from a cancer cell.

Knowledge Check Question

Question: Which cells normally make PTH?

Answer: The chief cells of the parathyroid glands normally make PTH.

How Does Parathyroid Hormone Act on Tissues?

PTH does its job by acting directly on bones, cartilage, and kidneys and acting indirectly on the gut.

Actions on Bones and Cartilage

PTH secretion is stimulated by hypocalcemia. The hormone **increases serum calcium** levels by stimulating osteoclasts in bones. Osteoclasts cause bone resorption, releasing bone calcium into the blood (Figure 4).

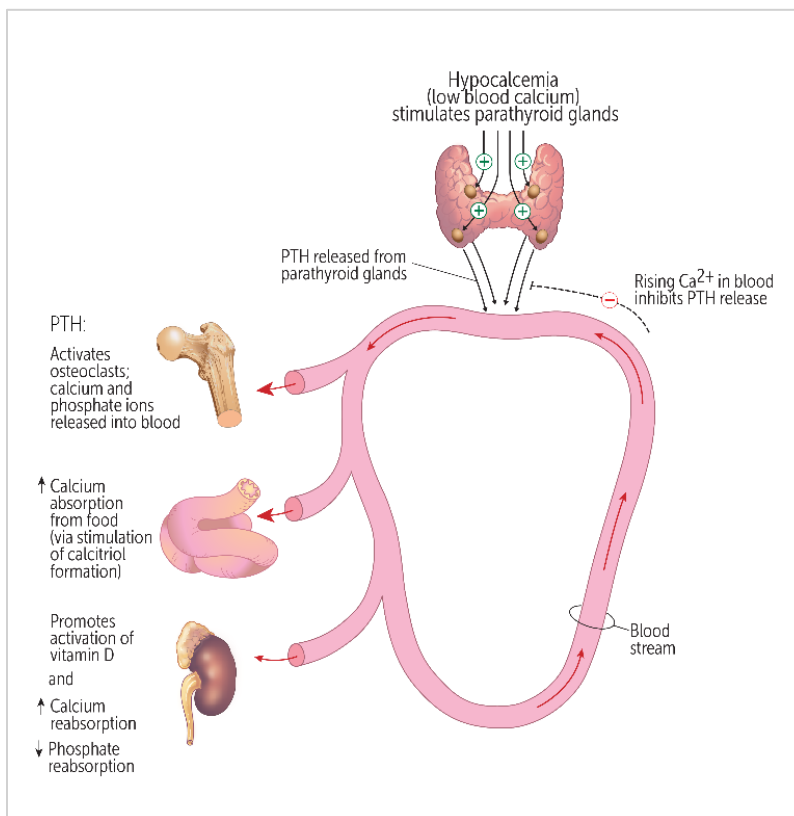


Figure 4 Credit: ©ScholarRx

This PTH stimulation of osteoclasts does not happen directly, because osteoclasts do not have PTH receptors. Instead, PTH stimulates bone-forming osteoblasts, which then activate osteoclasts by mechanisms involving cytokine secretion and the receptor activator of nuclear factor κ -B (RANK). The osteoclasts dissolve bone hydroxyapatite, releasing calcium into the bloodstream.

All these bone effects are opposed by the hormone **calcitonin**, released by C cells within the thyroid gland. Calcitonin inhibits osteoclasts and lowers the serum calcium concentration.

PTH also stimulates the proliferation of chondrocytes, which make cartilage. It is under investigation for use in treating patients with cartilage defects.

Knowledge Check Question

Question: What cells are activated by PTH in the bones to increase serum calcium levels?

Answer: Osteoclasts are activated by PTH in the bones to increase serum calcium levels. Activation of osteoclasts by PTH occurs indirectly via osteoblasts.

Actions on the Kidney

PTH increases renal calcium reabsorption and renal phosphate excretion. It directly stimulates calcium reabsorption in the thick ascending loop of Henle and in the cortical collecting duct. It also decreases phosphate reabsorption in the proximal tubule.

In addition, PTH stimulates renal 1- α -hydroxylase, which converts 25-hydroxy vitamin D to **1,25-dihydroxy vitamin D**, also called calcitriol. This is the active form of vitamin D. In turn, calcitriol increases intestinal and renal calcium reabsorption. Calcitriol can also act on bone to stimulate bone osteoclasts, but this is not a major function of calcitriol. A more important effect of calcitriol in bone is to make dietary calcium and phosphate available for bone formation.

The net effect of PTH activity is to increase serum calcium, decrease serum phosphate, and decrease bone calcium.

Actions on the Gut

PTH acts indirectly on the gut by stimulating formation of calcitriol, which then increases intestinal calcium absorption. The gut itself does not have PTH receptors.

Clinical Correlation



High PTH levels (hyperparathyroidism) may be caused by parathyroid adenomas. This leads to hypercalcemia, usually without symptoms. About 50% of patients will also have hypophosphatemia due to increased renal excretion.

How Is Parathyroid Hormone Secretion Regulated?

PTH secretion by the parathyroid gland is mainly regulated by the serum calcium level and calcitriol via negative feedback.

Serum Calcium Level

Low levels of serum calcium (hypocalcemia) stimulate PTH secretion, whereas high levels (hypercalcemia) suppress it. This makes sense, when you remember that PTH will act to increase the calcium level by releasing it from the bones and reabsorbing it from the kidney.

Parathyroid cells detect the serum calcium level with a sensitive **calcium-sensing receptor** on the cell membrane. The binding of calcium to the receptor causes the activation of guanine nucleotide-binding (G) proteins, which inhibit PTH secretion. The reverse happens when calcium levels are low—PTH is released by exocytosis.

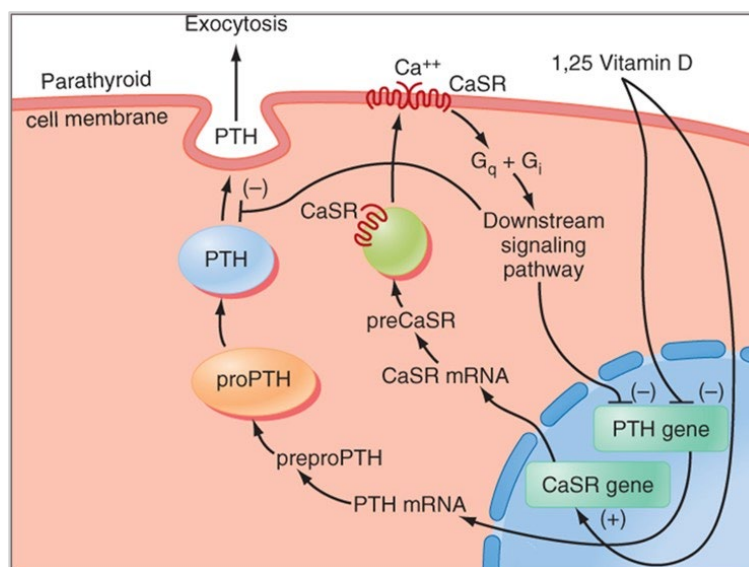


Figure 5: Regulation of PTH gene expression and secretion Credit: ©ScholarRx

Hypocalcemia is the primary stimulus for PTH release. Extracellular Ca^{2+} concentration is sensed by Ca^{2+} -sensitive receptor (CaSR) in chief cells of the parathyroid gland. These Gi/Gq GPCRs are very sensitive to changes extracellular calcium.

- Increased extracellular Ca^{2+} concentration $\rightarrow$ CaSR activation $\rightarrow$ $\uparrow$ intracellular Ca^{2+} and $\downarrow$ cAMP $\rightarrow$ $\downarrow$ PTH synthesis & release.
- Decreased extracellular Ca^{2+} concentration $\rightarrow$ CaSR “relaxed” $\rightarrow$ $\downarrow$ intracellular Ca^{2+} and $\uparrow$ cAMP $\rightarrow$ $\uparrow$ PTH synthesis & release.



Clinical Correlation



Patients with hyperparathyroidism may be treated with cinacalcet. This is a calcimimetic agent; by binding to the PTH receptor, the agent “fools” the receptor into “thinking” it’s been bound by calcium. This activates the receptor and decreases PTH secretion.

Calcitriol

Calcitriol (1,25-dihydroxy vitamin D) decreases PTH synthesis and release. It crosses the cell membrane and binds to the intracellular vitamin D receptor, which then transits to the nucleus to lower PTH gene expression.

Serum Phosphate Level

High serum phosphate levels, as seen in severe chronic kidney disease, stimulate PTH secretion. This is likely because extracellular phosphate inhibits the calcium-sensing receptors on the parathyroid gland. Inhibiting these receptors stimulates PTH secretion by the gland.

Clinical Correlation



This effect of phosphate on PTH is likely not important except in patients with very high phosphate levels, as is seen in patients with end-stage kidney disease. These patients have severe hyperphosphatemia because phosphate filtration in their kidneys is limited. As a result, these patients may develop very high PTH levels, called secondary hyperparathyroidism.

Knowledge Check Question

Question: What effects do hypocalcemia and hypercalcemia have on PTH?

Answer: Hypocalcemia stimulates PTH secretion and hypercalcemia inhibits it.



Case Connection

Thinking back to SF, how would you answer her question regarding where parathyroid hormone exerts its effects?

“Parathyroid hormone is a great multitasker,” you begin. “In addition to working on the bones, parathyroid hormone acts on the kidney and the gut to increase the absorption of calcium. When we remove your parathyroid glands, the hormone level will return to a normal range and your calcium level will also normalize. All that multitasking stops.” “Sounds so complicated but yet so simple,” SF says. “I’m so glad you understand all of this and can explain it to me so I can understand. I think that is such an important skill. Thanks for taking the time.”

Summary

- The parathyroids are four small glands that secrete parathyroid hormone (PTH), which helps regulate calcium and phosphate.
- Parathyroid glands are variable in location and in number.
- Most people have four parathyroid glands, which can be found on the posterior surface of the thyroid.
- The parathyroid glands receive their blood supply from the inferior thyroid artery.
- The parathyroid glands drain to the superior, middle, and inferior thyroid veins.
- PTH is a peptide hormone secreted as a prohormone.
- PTH binds to parathyroid hormone receptors, which are G protein-coupled receptors.
- PTH binds to bone osteoblasts, which activate adjacent osteoclasts to resorb bone and release calcium into the circulation.
- PTH stimulates renal calcium reabsorption and phosphate excretion.
- PTH stimulates formation of calcitriol, which also acts to increase the serum calcium.
- PTH secretion is stimulated by hypocalcemia and inhibited by hypercalcemia.
- PTH secretion is inhibited by calcitriol.
- PTH secretion is increased by the high serum phosphate concentrations seen in chronic kidney disease.



Check Your Knowledge

Part 1 Structure and Function of the Parathyroid Glands Review Questions

Question 1

The parathyroid glands are located on which surface of the thyroid gland?

- A. Anterior aspect
- B. Inferior aspect
- C. Lateral aspect
- D. Posterior aspect
- E. Superior aspect

Question 2

Which of the following arteries supplies the parathyroid glands?

- A. External carotid artery
- B. Inferior thyroid artery
- C. Internal carotid artery
- D. Superior thyroid artery

Question 3

A 20-year-old female is found to have an elevated serum parathyroid hormone (PTH) level during evaluation of bone pain. Which of the following is most likely to stimulate PTH release?

- A. High 1,25-dihydroxy vitamin D level
- B. Hypercalcemia
- C. Hypocalcemia
- D. Hypophosphatemia
- E. Low calcitonin levels

Question 4

Which of the following is a likely effect of elevated parathyroid hormone?

- A. Activation of osteoclast parathyroid hormone receptors
- B. Decreased chondrocyte proliferation
- C. Hypocalcemia
- D. Hypophosphatemia
- E. Increased renal calcium excretion



Part 1 Structure and Function of the Parathyroid Glands Review Questions and Answers

Question 1

The parathyroid glands are located on which surface of the thyroid gland?

- A. Anterior aspect
- B. Inferior aspect
- C. Lateral aspect
- D. Posterior aspect**
- E. Superior aspect

Feedback: The correct answer is posterior aspect (D). The parathyroid glands are located on the posterior aspect of the thyroid gland. They are usually four in number, with two in each lobe of the thyroid gland. They occupy the upper and lower poles of each thyroid lobe and are hence called superior and inferior parathyroid glands. The other answer choices—anterior aspect (A), inferior aspect (B), lateral aspect (C), and superior aspect (E)—are incorrect.

Question 2

Which of the following arteries supplies the parathyroid glands?

- A. External carotid artery
- B. Inferior thyroid artery**
- C. Internal carotid artery
- D. Superior thyroid artery

Feedback: The correct answer is inferior thyroid artery (B). The parathyroid glands are supplied by the inferior thyroid artery, which branches off from the thyrocervical trunk. The external carotid artery (A) branches into the superior thyroid artery (D), which plays a major role in the blood supply of the thyroid gland, whereas the internal carotid artery (C) does not have any branches in the neck.

Question 3

A 20-year-old female is found to have an elevated serum parathyroid hormone (PTH) level during evaluation of bone pain. Which of the following is most likely to stimulate PTH release?

- A. High 1,25-dihydroxy vitamin D level
- B. Hypercalcemia
- C. Hypocalcemia**
- D. Hypophosphatemia



E. Low calcitonin levels

Feedback: The correct answer is hypocalcemia (C). Hypocalcemia stimulates parathyroid hormone (PTH) secretion by the parathyroid glands because not enough calcium ions bind to the PTH receptors. A high 1,25-dihydroxy vitamin D level (A) reduces PTH secretion, as do hypercalcemia (B) and hypophosphatemia (D). Low calcitonin levels (E) would raise the serum calcium level, which would inhibit PTH release.

Question 4

Which of the following is a likely effect of elevated parathyroid hormone?

- A. Activation of osteoclast parathyroid hormone receptors
- B. Decreased chondrocyte proliferation
- C. Hypocalcemia
- D. Hypophosphatemia**
- E. Increased renal calcium excretion

Feedback: The correct answer is hypophosphatemia (D). Parathyroid hormone (PTH) lowers serum phosphate levels by increasing renal phosphate excretion, mostly by blocking phosphate reabsorption in the proximal tubule. PTH cannot cause activation of osteoclast parathyroid hormone receptors (A), because that cell type does not have them (osteoblasts do). PTH stimulates chondrocyte hyperplasia and proliferation, so decreased chondrocyte proliferation (B) is incorrect. PTH releases calcium from bones, causing hypercalcemia, not hypocalcemia (C). It does this by increasing calcium reabsorption in the kidney, decreasing calcium excretion, so increased renal calcium excretion (E) is incorrect.



Function and Disorders of the Parathyroid Gland Part 2 of 4: Hypocalcemia and Hypercalcemia

Overview

Unless otherwise noted, all content and images for this lecture are adapted from ScholarRx and used with permission.

Learning Objectives

This page covers the following learning objectives:

Physiology

4. Define hypocalcemia and the normal calcium reference range.
5. Define hypercalcemia and the normal calcium reference range.
6. Describe the physiologic effects of hypocalcemia on nerve and muscle cells and cardiac conduction.
7. Describe the physiologic effects of hypercalcemia on nerves, muscle cells, and cardiac conduction.

Pathology

3. Understand the pathogenesis of 'hypercalcemia in malignancy. Differentiate the role of PTHrP from PTH in malignancy.

Clinical Medicine

2. Discuss the differential diagnoses for hypercalcemia and hypocalcemia.
3. Compare and contrast different clinical treatments for hypercalcemia.

Pharmacology

1. Explain the pharmacological treatment options for hypocalcemia, including the mechanisms of action of calcium supplements and vitamin D analogs.
2. Evaluate the appropriate use of intravenous calcium in managing severe hypocalcemia.
3. Discuss the pharmacological management of hypercalcemia, including the roles of bisphosphonates, calcitonin, furosemide, and gallium nitrate.

Hypocalcemia

Case Connection

JC is a 42-year-old receptionist who had gastric bypass surgery 2 years ago. She has lost more than 90 lb. She was feeling well until a few weeks ago, when she had some pain in her chest. In the emergency department, she was diagnosed with esophageal spasms and sent home with a plan to return if it happened again. Although it has not happened again, you noticed that her ECG from the emergency department visit showed a prolonged QT interval. You ask whether she has been taking the prescribed supplements, and she admits she stopped taking them about 6 months ago. "It was such a chore! I eat a pretty good diet, so I thought I was getting what I needed anyway."

Is there a connection between JC's ECG abnormalities and her prior surgery? Consider your answer as you read, and we'll revisit JC at the end of the brick.

What Is Hypocalcemia?

As implied by its name, hypocalcemia is a reduced concentration of calcium in the blood. The reference range for total serum calcium is 8.4 to 10.2 mg/dL, so a value less than 8.4 mg/dL would be considered hypocalcemia. If it happens acutely, it can lead to unpleasant muscular symptoms (twitching, cramping), which can worsen into cardiac abnormalities and seizures in more severe cases.

How Do Patients With Hypocalcemia Present?

The signs and symptoms of hypocalcemia depend on the rapidity of its onset and severity of the calcium deficiency.

- Most hypocalcemia is **chronic and asymptomatic**.
- Hypocalcemia frequently is asymptomatic. However, in general, the more rapid the onset and the more severe the deficiency, the more likely one is to experience symptoms. This makes **acute hypocalcemia** much more dangerous and potentially life-threatening.

Fortunately, there is a single common underlying mechanism behind the clinical presentations of hypocalcemia: when calcium levels are low, it changes the transmembrane electrochemical gradient. This alters the permeability of cellular membranes to sodium and changes the resting membrane potential. Basically, it lowers the threshold for depolarization, making the nervous system and cardiac cells hyperexcitable.

The effects of this change manifest most prominently in neurons and cardiac cells, which are under constant, synchronized electrical control. Understanding that the heart and neurons are most likely

to be affected by hypocalcemia, we can understand the three general categories of the symptoms of hypocalcemia: neuromuscular irritability, psychiatric disturbances, and cardiac instability (see Complications, below).

Neuromuscular Irritability

Neuromuscular irritability refers to the hyperexcitability of neurons. In hypocalcemia, because of the altered membrane permeability, neurons fire unpredictably. This presents in numerous ways.

Paresthesias, or tingling sensations (pins and needles), are a common initial symptom that patients feel, especially around the mouth and in the fingers and toes. **Muscle twitching** and **hyperreflexia** are common as well, as you might expect.

Two clinical signs of neuromuscular irritability from hypocalcemia that clinicians often elicit when they suspect hypocalcemia are the Chvostek sign and Trousseau sign (Figure 1).

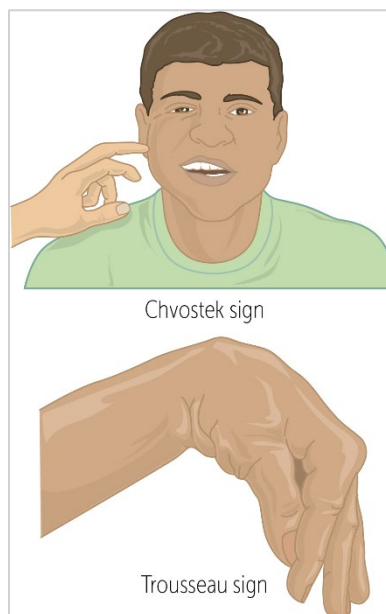


Figure 1 Credit: ©ScholarRx

How do we elicit these signs?

- **Chvostek sign** occurs when tapping the facial nerve leads to twitching of the facial muscles.
- **Trousseau sign** is elicited by inflating a blood pressure cuff to a pressure higher than the patient's systolic blood pressure for three minutes. If this leads to spasm and contracture of the wrist and carpal muscles, then the sign is considered positive.

As hypocalcemia worsens and neurons become even more irritable, potentially fatal symptoms such as laryngospasm, bronchospasm, and seizures can occur. As we will see, acute hypocalcemia can also affect cardiac conduction, leading to arrhythmias.

Knowledge Check Question

Question: What are three neuromuscular signs and symptoms of acute hypocalcemia?

Answer: Three neuromuscular signs and symptoms of acute hypocalcemia are hyperreflexia, paresthesias, and muscle twitching (also Chvostek sign, Trousseau sign, bronchospasm, laryngospasm, and seizures).

Psychiatric Disturbances

When neurons in the brain are irritable and unstable, psychiatric disturbances may present as well. These disturbances include anxiety and depression. Altered mental status (confusion, memory impairment, and decreased consciousness) and psychosis are also possible.

Clinical Correlation



Chronic hypocalcemia can lead to calcification of the basal ganglia and parkinsonism. It also can cause cataracts and dementia.

Knowledge Check Question

Question: When thinking of the clinical symptoms of hypocalcemia, think “jumpy, excited neurons.” This is the underlying theme, and it will help you remember most important clinical manifestations.

Answer: Clinical symptoms of hypocalcemia: jumpy, excited neurons

What is the Pathophysiology of Hypocalcemia?

There are many possible causes of hypocalcemia. These will have a variety of effects on the body, some depending on how rapid the fall in calcium is. A common consequence is neuromuscular excitability, and we will discuss the mechanism for that at the end of this section.

Etiology

The main causes include hypoparathyroidism, chronic kidney disease and hyperphosphatemia, vitamin D deficiency, and medication side effects.

Hypoparathyroidism

The most common cause of hypocalcemia is hypoparathyroidism, which is characterized by low levels of parathyroid hormone (PTH) in the blood. The function of PTH is to increase serum calcium levels, and it does this through its actions on the bones, kidneys, and gastrointestinal (GI) tract. The most common cause of hypoparathyroidism is surgery to the thyroid gland that damages or removes one or more of the parathyroid glands. Other less common causes include autoimmune destruction of the parathyroid glands or DiGeorge syndrome, in which the parathyroid glands fail to develop.

Chronic Kidney Disease and Hyperphosphatemia

Patients with advanced chronic kidney disease (CKD) may develop hypocalcemia for two reasons. First, damage to the renal tissue leads to inadequate activation of vitamin D to calcitriol, which normally increases serum calcium. Second, high serum phosphate in CKD precipitates with calcium to create deposits of calcium phosphate. This removes elemental calcium from the serum, resulting in hypocalcemia. The low serum calcium drives secondary hyperparathyroidism and chronic bone disease in these patients with CKD.

Vitamin D Deficiency

Vitamin D deficiency is another common cause of hypocalcemia, especially in developing nations where there is lower access to dairy products or supplemented foods. Normally, vitamin D acts to increase serum calcium levels by increasing calcium absorption in the GI tract. Without enough vitamin D, calcium absorption can be impaired, leading to hypocalcemia. The many ways someone can become vitamin D deficient include insufficient dietary intake, decreased sun exposure, gut malabsorption (eg, bariatric surgery), liver disease (failure of 25-hydroxylation of vitamin D), and chronic kidney disease. Individuals with increased skin pigmentation are also at higher risk for vitamin D deficiency.

Hypomagnesemia

Magnesium is necessary for the proper secretion of PTH and response to PTH by target tissues. Low magnesium may result in both decreased PTH secretion and reduced renal responsiveness to PTH, leading to hypocalcemia. Hypomagnesemia is often seen in people with dietary deficiencies, such as chronic alcohol users. In patients with hypocalcemia with low magnesium, the serum calcium will not be corrected until the magnesium is corrected.

Gastrointestinal Malabsorption

The underlying inability to absorb calcium in the small intestine is another cause of hypocalcemia. Surgery to the GI tract that shortens the length of, or otherwise damages, the intestine can lead to the malabsorption of numerous nutrients, vitamins, and minerals, calcium among them.

Blood Transfusion

Transfused blood typically contains citrate, which helps prevent stored blood from clotting. An undesirable effect of this citrate is that it can precipitate calcium out of the serum, resulting in severe, acute hypocalcemia. This can have severe effects such as seizures.



Acute Pancreatitis

Hypocalcemia is a complication of acute pancreatitis. The exact mechanism is unknown. It may be caused by the high serum lipids seen after the digestion of adipose tissue by pancreatic enzymes released into the blood. These lipids form soaps with the serum calcium, reducing the level of ionized calcium. Hypomagnesemia may also play a role.

Medications

Loop diuretics (eg, furosemide) can increase renal calcium excretion, leading to hypocalcemia. Bisphosphonates (eg, zoledronate) can cause hypocalcemia because they bind to calcium in the serum or bone. Cinacalcet mimics calcium at the parathyroid gland calcium receptors. This action results in decreased PTH secretion and can lead to hypocalcemia. Foscarnet, an antiviral drug, forms complexes with free ionized calcium and can cause hypocalcemia.

Knowledge Check Question

Question: What is the most common cause of hypocalcemia?

Answer: Hypoparathyroidism is the most common cause of hypocalcemia.

Complications: Nerve Conduction and Cardiac Instability

As noted, hypocalcemia causes a variety of neuromuscular effects, such as tetany and seizures. How does reduced serum calcium affect the cells? Normally, calcium blocks sodium channels and inhibits the depolarization of nerves and muscles. When there is lower serum calcium, less sodium is blocked, causing the threshold potential (depolarization threshold) to fall (become more negative).

As shown in Figure 2, acute hypocalcemia decreases the distance between the resting potential and threshold potential, making depolarization more likely and less controlled. This explains the “hyper” and twitch neurologic effects of hypocalcemia.

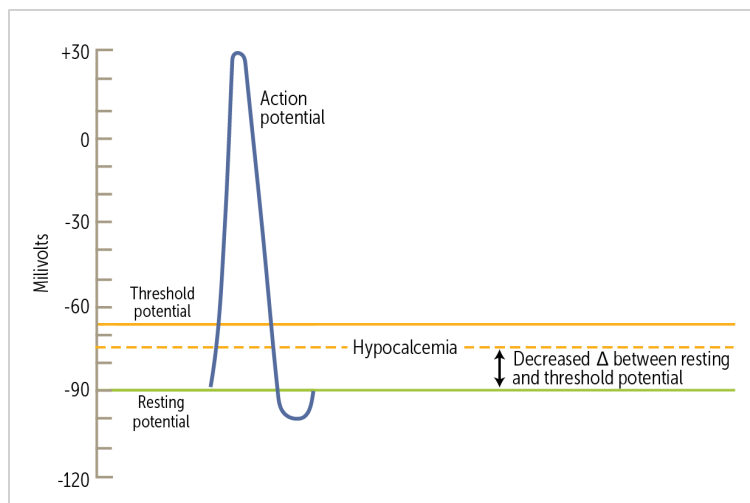


Figure 2 Credit: ©ScholarRx

How might this affect the heart? The heart is under highly controlled, synchronized electrical control, so alterations in permeability to ions can have drastic effects on it. One potentially dangerous conduction abnormality associated with hypocalcemia is an ECG showing **prolonged QT interval**, which you can see compared with a normal QT interval in Figure 3.

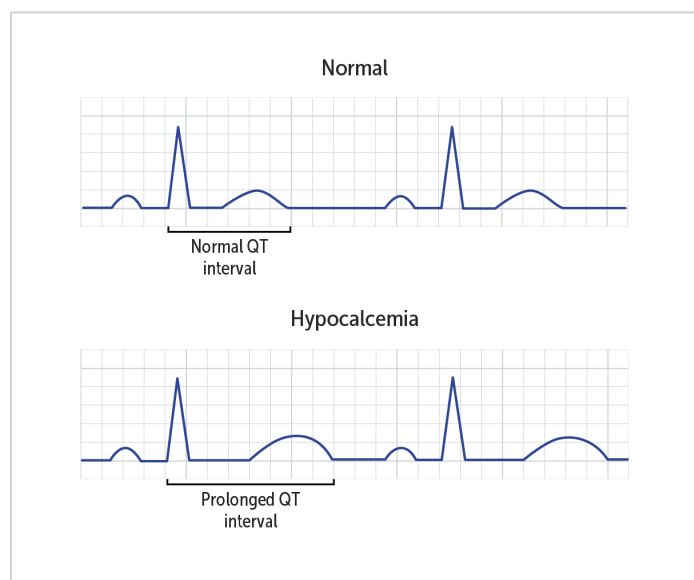


Figure 3 Credit: ©ScholarRx

The QT interval is prolonged because ventricular depolarization is prolonged. This is caused by the **prolongation of the plateau phase** of the cardiac myocyte action potential. The prolonged plateau causes calcium ion channels to remain open for a longer period, allowing for a late calcium inflow. Conduction abnormalities, arrhythmias, and decreased force of contraction may result. These usually only occur with severe hypocalcemia.

A prolonged QT interval is especially dangerous because it can lead to the potentially fatal arrhythmia known as **torsades de pointes** (Figure 4).

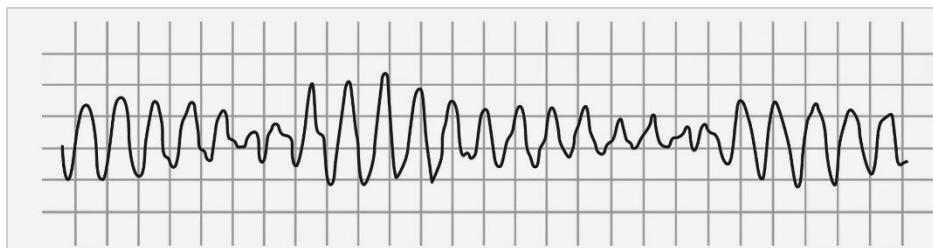


Figure 4 Credit: ©ScholarRx

Knowledge Check Question

Question: What ECG abnormality is often a result of hypocalcemia?

Answer: QT prolongation

How Do We Diagnose Hypocalcemia?

As described above, the Chvostek sign and Trousseau sign are specific findings on physical exam that would point toward a diagnosis of hypocalcemia. On suspicion of hypocalcemia, laboratory tests are used to confirm a diagnosis.

Measuring Serum Calcium

The first and most important test is to measure the total serum calcium level. If the patient has suspected low serum albumin (eg, a patient with cirrhosis or kidney disease), measure the serum albumin so you can correct the serum calcium. The total calcium can be adjusted for changes in serum albumin with this formula:



$$\text{corrected Ca}^{2+} = \text{measured total Ca}^{2+} + (0.8 \times [4.0 - [\text{albumin}]])$$

Another option is to order an ionized calcium level to avoid having to make the correction and run an albumin level test separately.

Auxiliary Testing

The other important tests to perform on a patient with documented hypocalcemia focus on identifying the underlying causes discussed above. These include serum PTH, phosphate, magnesium, vitamin D, and creatine levels (Table 1).



Table 1

	PTH	Phosphate	Magnesium	Vitamin D	Creatine
Hypoparathyroidism	Low	Elevated	Normal	Normal or low	Normal
Hypomagnesemia	Low, normal, or high	Normal	Low	Normal	Normal
Vitamin D deficiency	Elevated	Low or normal	Normal	Low	Normal
Chronic kidney disease	Elevated	Elevated	Low	Normal or low	Elevated

Electrocardiogram

Additionally, it is important to do an ECG in the setting of hypocalcemia. With suspicion of hypocalcemia, this can be done before labs results have arrived. The hallmark of hypocalcemia on ECG is prolongation of the QT segment secondary to prolongation of the ST segment. It is important to monitor the QT interval prolongation (see Figure 3) because it predisposes patients with hypocalcemia to severe arrhythmias such as torsades des pointes, a form of ventricular fibrillation.

Knowledge Check Question

Question: What two physical exam findings are specific for hypocalcemia?

Answer: Chvostek sign and Trousseau sign are two physical exam findings that are specific for hypocalcemia.

How Do We Manage Hypocalcemia?

Hypocalcemia usually results from a different disease, so the underlying cause (eg, hypomagnesemia, hyperphosphatemia) should be sought and corrected, if possible.

Asymptomatic Patients

Asymptomatic hypocalcemia may be treated with oral calcium supplementation (eg, calcium carbonate).

Symptomatic Patients

Symptomatic hypocalcemia or hypocalcemia with correlated ECG changes (long QT interval) is dangerous and requires immediate treatment with **intravenous (IV) calcium gluconate** to rapidly correct calcium levels. A patient with symptomatic hypocalcemia has unstable neurons and cardiac cells, and the purpose of this treatment is to restore normal cardiac function and prevent bronchospasm and seizures—all of which can be acutely fatal.

Depending on the etiology, some patients may require long-term oral calcium supplementation, along with vitamin D to facilitate its absorption.



Knowledge Check Question

Question: What is the treatment for acute symptomatic hypocalcemia?

Answer: IV calcium gluconate is the treatment for acute symptomatic hypocalcemia.

Clinical Correlation

Thinking back to JC, is there a connection between her ECG abnormalities and her prior surgery?

Because of the postsurgical changes to JC's gut anatomy, she is at high risk of malabsorption of both calcium and vitamin D (as well as other nutrients). Concerned about possible deficiencies, you order a comprehensive metabolic panel. Her calcium level is low. You remember that you need to correct for albumin. Her corrected calcium is still low. You repeat the ECG, and the QT prolongation remains. You explain to JC that it is important to rapidly replace the calcium to avoid a dangerous heart rhythm. You admit her to the hospital to receive IV calcium gluconate and arrange for the bariatric dietitian to meet with her.

Summary of Hypocalcemia

- Hypocalcemia is a serum concentration of calcium ions that is lower than the normal range of 8.4-10.2 mg/dL.
- Hypocalcemia causes hyperexcitability of neurons and cardiac cells by altering their permeability to sodium ions and changing the resting membrane potential.
- Signs and symptoms of hypocalcemia include muscle spasms, twitching, paresthesias, Chvostek and Trousseau signs, prolonged QT, and psychiatric disturbances.
- Hypocalcemia has many etiologies, such as hypoparathyroidism (most common) vitamin D deficiency, malabsorption, and chronic kidney disease.
- Loop diuretics and bisphosphonates can cause hypocalcemia.
- The lab evaluation for suspected hypocalcemia should include measurement of total or ionized serum calcium and vitamin D, as well as the evaluation of possible underlying causes.
- Acute symptomatic hypocalcemia requires treatment with IV calcium gluconate.
- Long-term management of hypocalcemia requires oral calcium supplements, vitamin D supplements, and treatment of any underlying causes.

Hypercalcemia

Case Connection

TG is a 67-year-old male, who formerly smoked. He came in with cough and abdominal pain and has just been diagnosed with lung cancer. He was planning to retire at the end of the year but wonders whether he should do it sooner because he is struggling to get through a workday. He feels tired, and his bones hurt. You review his lab results and note an elevated serum calcium level. “What about the pain in his bones? Does that mean the cancer has already spread?” his wife asks.

How will you explain the bone pain to TG and his wife? Consider your answer as you read, and we’ll revisit TG at the end of the brick.

What Is Hypercalcemia?

Hypercalcemia is defined as having too much calcium in your blood. The typical calcium serum reference range is 8.4 to 10.2 mg/dL, so 10.3 mg/dL or greater is considered hypercalcemia. As we’ll see, hypercalcemia can be due a range of problems, including endocrine issues (high parathyroid hormone), cancer, even immobility. And the symptoms range widely, from none to marked confusion and metabolic abnormalities. It’s a good test of your diagnostic skills, so let’s investigate further.

How Do Patients With Hypercalcemia Present?

Signs and symptoms of hypercalcemia vary depending on disease severity and how fast the disease arises:

- Most cases of hypercalcemia are **chronic and asymptomatic**. This form is most often caused by primary hyperparathyroidism.
- **Acute** hypercalcemia is generally symptomatic when the calcium level exceeds 12 mg/dL and is most often seen in patients with cancer.

When symptoms do exist, they can appear anywhere in the body that’s affected by high serum calcium levels, notably the nervous system, gastrointestinal tract, kidney, bones, and skeletal muscles.

Nervous System

Patients with hypercalcemia may exhibit central nervous system or psychiatric effects. These include **lethargy, confusion, or even coma** when levels exceed 14 mg/dL. Anxiety and depression are described in patients with chronic hypercalcemia.

Gastrointestinal System

Hypercalcemia may cause **constipation**, anorexia, vomiting, and acute pancreatitis. High serum calcium can increase the level of gastrin, which increases gastric acid production. Ultimately, this may lead to peptic ulcer disease or gastritis.

Urinary System

Chronically high urine calcium levels may lead to calcium kidney stones, causing **flank pain, groin pain, and hematuria**.

Hypercalcemia is also a cause of **polyuria** and nephrogenic diabetes insipidus, because high calcium inhibits the action of antidiuretic hormone in the renal tubules. The high urine output associated with diabetes insipidus will also cause polydipsia (abnormal increase in thirst).

Hypercalcemia can also cause distal renal tubular acidosis (type 1 RTA) and acute kidney injury.

Musculoskeletal System

Chronic hypercalcemia (resulting from hyperparathyroidism) is associated with **mild muscle weakness**, bone pain, osteopenia, and fractures. These effects are usually caused by an increase in osteoclast activity.

Findings from these systems are sometimes summarized (in no particular order) as moans (nervous system), groans (digestive system), stones (urinary system), and bones (musculoskeletal) (Figure 1).

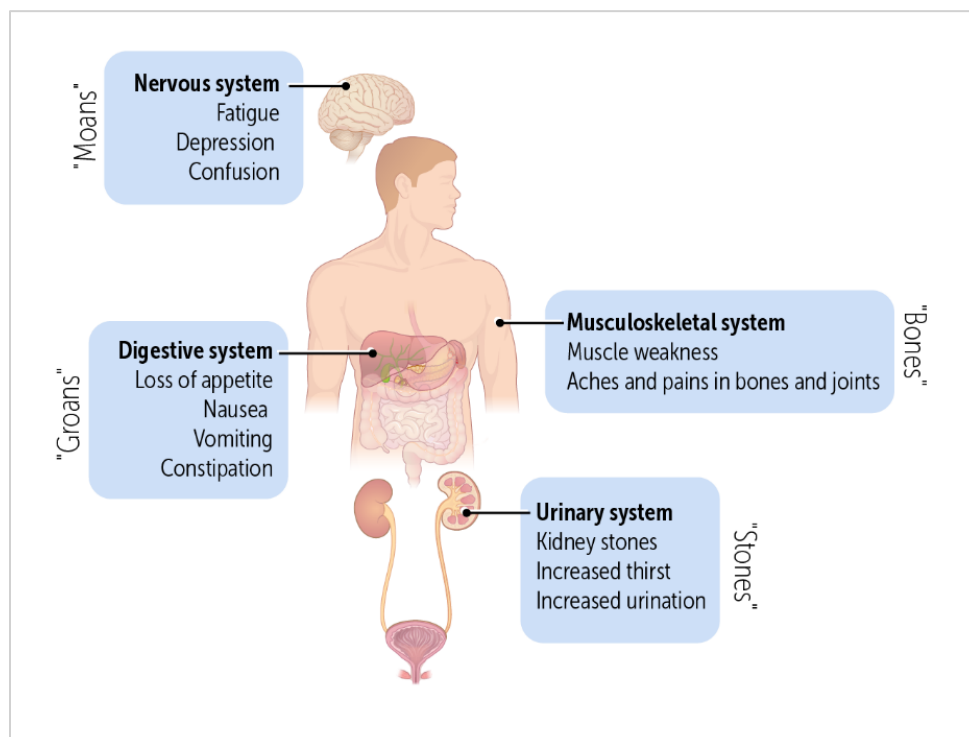


Figure 1 Credit: ©ScholarRx

Knowledge Check Question

Question: Clinical symptoms of hypercalcemia:

- Stones, bones, groans, and moans.
 - Stones (nephrolithiasis)
 - Bones (bone aches and pains, fractures, osteitis fibrosa cystica)
 - Groans (muscle pain, constipation, anorexia, peptic ulcer disease)
 - Moans (depression, fatigue, sleep disturbances, lethargy)
-

Answer: Clinical symptoms of hypercalcemia: Stones, bones, groans, and moans

Knowledge Check Question

Question: Patients with what condition most often present with acute hypercalcemia?

Answer: Patients with cancer most often present with acute hypercalcemia.

What Is the Pathophysiology of Hypercalcemia?

Hypercalcemia develops when calcium absorption from the diet or resorption from the bones exceeds urinary calcium excretion. This can affect muscle contraction and nerve conduction.

Etiology

Several conditions disrupt the balance of calcium in the body and result in hypercalcemia, including hyperparathyroidism, cancer, and granulomatous diseases.

Hyperparathyroidism

Primary hyperparathyroidism is the most common cause of hypercalcemia. Certain conditions cause one or more of the parathyroid glands to become overactive, leading to excess production of parathyroid hormone (PTH) and hypercalcemia. The most common cause of primary hyperparathyroidism is a benign adenoma. Other causes of primary hyperparathyroidism include parathyroid hyperplasia (diffuse enlargement of all four glands) and, more rarely, parathyroid carcinoma and MEN syndromes.

Cancer

Hypercalcemia is common in patients with cancer, particularly in those with advanced disease. Approximately 25% of malignancies will cause hypercalcemia at some point during the course of the disease. Malignancy-associated hypercalcemia generally arises as a consequence of one of the following: production of parathyroid hormone-related protein (PTHrP) by cancer cells, metastatic



invasion of bone, or production of the enzyme 1α -hydroxylase. Let's take a look at each of these mechanisms:

- **PTHrP:** Some tumors (eg, breast and lung carcinoma) produce a protein similar in structure to PTH and secrete it into systemic circulation. PTHrP can bind to and activate the same receptors, leading to hypercalcemia. PTHrP can be measured, but it is not detected in the conventional PTH lab assays.
- **Metastatic invasion of bone:** Sometimes when cancers metastasize (or spread) to bone, they disrupt the balance between bone formation and resorption. Metastatic invasion of bone often stimulates osteoclastic bone resorption and the subsequent release of calcium. This can increase serum calcium levels. Cancers that typically invade the bone include multiple myeloma, breast cancer, and lung cancer.
- **Increased formation of calcitriol:** Some lymphomas can produce extrarenal (ectopic) 1α -hydroxylase, which normally converts inactive vitamin D to its active form, calcitriol ($1,25$ -(OH) $_2$ vitamin D). Excess calcitriol increases serum calcium levels.

Granulomatous Disorders

A granuloma is an area of chronic inflammation with an aggregation of macrophages. Conditions that cause granuloma formation include sarcoidosis and tuberculosis. Macrophages in granulomas express 1α -hydroxylase, leading to excessive amounts of calcitriol and hypercalcemia. PTH levels are low.

Medications

Certain medications and supplements can also cause hypercalcemia. Thiazide diuretics (eg, chlorthalidone) increase renal reabsorption of calcium in the distal convoluted tubule and occasionally cause hypercalcemia. Lithium increases the release of PTH and can subsequently cause elevated serum calcium levels.

Excessive supplementation with calcium (milk-alkali syndrome) or vitamin D, often taken for bone health, can also lead to hypercalcemia.

Familial Hypocalciuric Hypercalcemia

Familial hypocalciuric hypercalcemia is a rare inherited disorder marked by persistent hypercalcemia (usually >10.2 mg/dL). It's usually asymptomatic. Most cases are caused by defects in the calcium-sensing receptors in the parathyroids and renal tubules. Familial hypocalciuric hypercalcemia is the only cause of hypercalcemia that shows low urine calcium excretion.

Figure 2 summarizes the effects of some of these disorders on the serum PTH and calcium levels, comparing them to hypoparathyroidism.

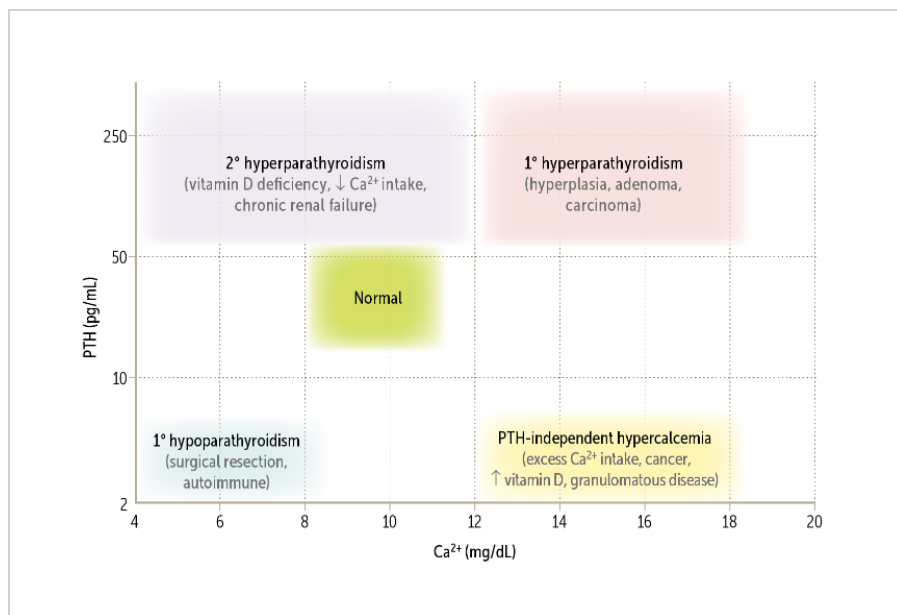


Figure 2 Credit: ©ScholarRx

Knowledge Check Question

Question: What is the most common cause of hypercalcemia?

Answer: Primary hyperparathyroidism is the most common cause of hypercalcemia.

Cardiac and Nerve Complications

How does elevated calcium affect the cells? Because calcium is an important component of many cell signaling pathways, elevated calcium can have diverse effects, including abnormalities in nerve conduction and in skeletal and cardiac muscle contraction. What's the mechanism?

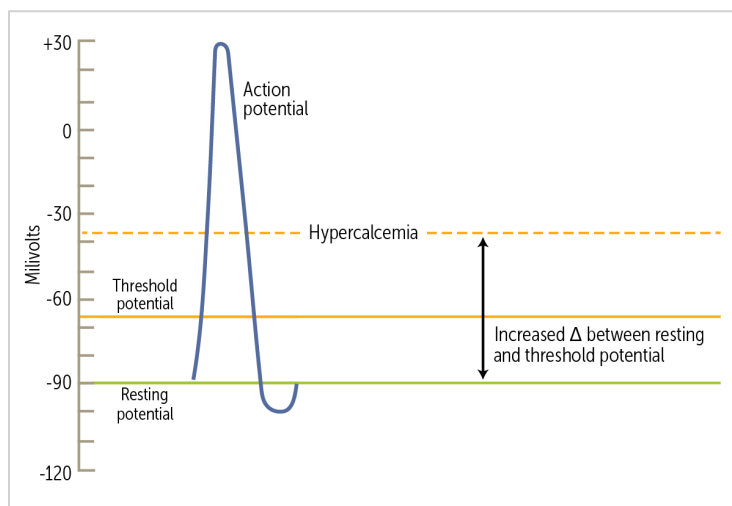


Figure 3 Credit: ©ScholarRx



Calcium blocks sodium channels and **inhibits depolarization** of nerves and cardiac and skeletal muscle. When there is extra calcium, more sodium is blocked, causing the threshold potential (depolarization threshold) to rise (become less negative). As shown in Figure 3, acute hypercalcemia increases the distance between the resting potential and the threshold potential (raising it from the solid line to the dotted line).

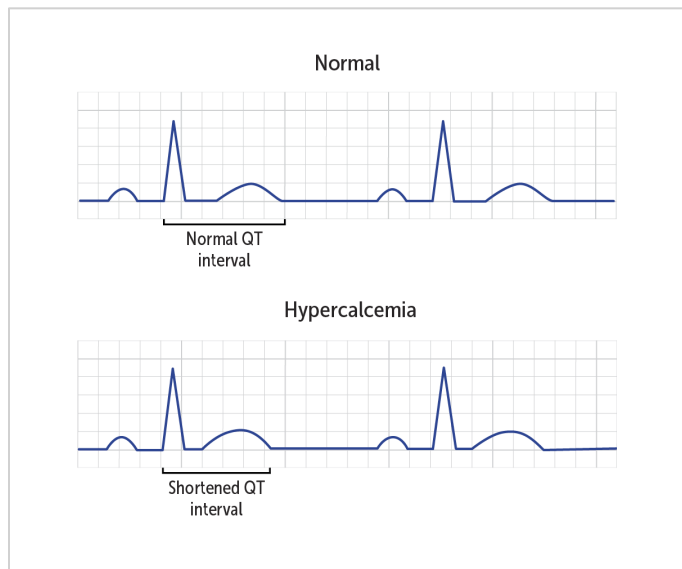


Figure 4 Credit: ©ScholarRx

This makes depolarizations less likely and explains many of the observed neurologic and cardiac effects of hypercalcemia, such as lethargy, constipation, and weakness.

What is the effect of hypercalcemia on the heart? The high calcium levels block sodium movement through voltage-gated sodium channels, slowing sodium entry into excitable membranes. The reduced depolarization of cardiac myocytes shortens cardiomyocyte repolarization time, so the **QT interval is reduced** (Figure 4).

In more severe hypercalcemia (usually over 14 mg/dL) this can lead to ST segment changes, which can appear like a myocardial infarction, and sometimes arrhythmias.

Longstanding hypercalcemia may cause calcium deposits on the heart valves, coronary arteries, and myocardial fibers, which can lead to valve stenosis, hypertension, and cardiomyopathy.

Knowledge Check Question

Question: What arrhythmia is the most commonly seen on ECGs for hypercalcemia?

Answer: Shortened QT interval is the arrhythmia most commonly seen on ECGs for hypercalcemia.



How Do We Diagnose Hypercalcemia?

Because hypercalcemia so often is asymptomatic, we rely on lab testing to determine if calcium levels are high. In fact, a diagnosis may result from a routine blood screening, such as one ordered as part of a patient's annual exam. After documenting high calcium, we need to determine its cause.

Measuring Serum Calcium

Calcium in our blood comes in two forms: calcium bound to albumin and free calcium in its ionized form. As the ionized (free) calcium level is the most clinically relevant measurement, it can be measured directly as an ionized calcium level. The total calcium can be adjusted for changes in serum albumin using this formula:



$$\text{corrected } \text{Ca}^{2+} = (0.8 \times [\text{normal albumin} - \text{patient's albumin}]) + \text{serum } \text{Ca}^{2+} \text{ level}$$

Normally, total calcium decreases by 0.8 mg/dL for every 1 g/dL decrease in serum albumin below 4 g/dL. Therefore, using the formula helps us detect hypercalcemia masked by low serum albumin.

Auxiliary Testing

After hypercalcemia is documented, the workup will depend on the most likely cause based on the history and physical. Testing may include serum parathyroid hormone, phosphate, vitamin D, and PTHrP levels. Serum calcitriol can be measured to test for granulomatous disease. Multiple myeloma can be confirmed with electrophoresis of urine and serum protein.

Urine calcium is often high, except in cases of thiazide diuretic use and familial hypocalciuric hypercalcemia; in both of those cases, urine calcium is low.

Table 1 summarizes the hypercalcemia workup.

	Calcium	Phosphorus	PTH	PTHrP	Notes
Hyperparathyroidism	↑	↓	↑	↓	
Cancer (PTHrP mediated)	↑	↓	↓	↑	
Cancer (bone metastasis)	↑	↑	↓	↓	Normal vitamin D levels
Granulomatous disease	↑	↑	↓	↓	↑ 1,25-(OH) ₂ vitamin D
Vitamin D intoxication	↑	↑	↓	↓	↑ 1,25-(OH) ₂ vitamin D
Familial hypocalciuric hypercalcemia	↑	↓	↑	↓	Low urine calcium levels

Electrocardiography

In addition, ECG may be a part of a hypercalcemia workup in the setting of cardiac symptoms. If a patient is complaining of chest pain or shortness of breath, for example, it is important to investigate the cause. ECG changes are common in the setting of hypercalcemia, and they may mimic patterns characteristic of myocardial ischemia. The most common finding is a **shortened QT interval**, mainly caused by shortening of the ST segment.

Knowledge Check Question

Question: Would you expect a patient with malignancy to have high or low serum PTH?

Answer: Patients with cancer have high PTHrP, which causes hypercalcemia. Hypercalcemia, in turn, lowers the PTH level.

How Do We Manage Hypercalcemia?

Treatment depends on whether the patient is asymptomatic or symptomatic.

Asymptomatic Patients

Patients with asymptomatic hypercalcemia do not require immediate action. However, thiazide diuretics, lithium carbonate therapy, volume depletion, prolonged bed rest or inactivity, and a high calcium diet (>1000 mg/d) should be avoided. The underlying cause should be found and treated (eg, cancer, hyperparathyroidism).

Symptomatic Patients

Significant symptoms usually develop in patients with calcium levels greater than 14 mg/dL, requiring immediate treatment to reduce the calcium concentration. Treatment goals of symptomatic hypercalcemia should be to immediately decrease the serum calcium concentration and resolve the root cause of the increased calcium level.

Volume Repletion

Most patients with severe hypercalcemia are volume depleted. The first treatment should always be volume expansion with intravenous fluids, to dilute the calcium in the serum and increase renal calcium excretion. Hypovolemia is corrected with 0.9% saline followed by adequate oral hydration.

Loop Diuretics

After volume depletion is corrected with saline, we can give loop diuretics to increase calcium excretion through the kidneys. Loop diuretics such as furosemide increase calcium excretion, in addition to their usual function to increase renal sodium and water excretion.

Knowledge Check Question

Question: “Loops lose calcium” is a helpful reminder that loop diuretics, such as furosemide, increase calcium excretion (vs thiazide diuretics that increase calcium retention).

Answer: Loops lose calcium

Bisphosphonates

For mild to moderate hypercalcemia that is resistant to fluids and loop diuretics, particularly in cases secondary to bone metastases, we can give bisphosphonates to inhibit bone calcium resorption, thereby preventing new calcium from being added to the serum. Bisphosphonates such as pamidronate, alendronate, and zoledronic acid are effective in limiting bone resorption by binding to hydroxyapatite, thereby preventing the osteoclast-mediated resorption that leads to pathologic fractures. The most common adverse effects are nausea and vomiting; esophageal erosions occasionally occur. Intravenous use has been associated with osteonecrosis of the jaw, especially after oral surgery procedures.

Calcitonin

If the patient has severe, acute hypercalcemia and a history of renal failure, calcitonin may be used. It is used in addition to the normal saline, particularly in severe cases when serum calcium levels increase above 14 mg/dL. Calcitonin inhibits osteoclast activity in bones, thereby decreasing bone resorption and calcium release. It also inhibits renal tubular reabsorption of calcium, increasing urinary calcium excretion. Calcitonin acts rapidly to reduce serum calcium concentration; however, calcitonin cannot be used long term because calcitonin resistance (tachyphylaxis) often develops with prolonged use. The most common adverse effects are nausea, vomiting, and diarrhea.

Knowledge Check Question

Question: Calcitonin tones down calcium concentration. Calcitonin inhibits osteoclast activity, thereby blocking the release of calcium into the serum.

Answer: Calcitonin “tones down” calcium concentration.

Other Agents

Denosumab

Denosumab is a monoclonal antibody that blocks the maturation of osteoclasts by binding to an intracellular signaling molecular called the RANK (receptor activator of nuclear factor κ) ligand; denosumab may be used to treat hypercalcemia of malignancy resistant to bisphosphonates.

Glucocorticoids

Glucocorticoids (eg, prednisone) are used inhibit the production of calcitriol by macrophages, thus treating hypercalcemia in granulomatous diseases, lymphoma, and tuberculosis.

Cinacalcet

Cinacalcet is a parasympathomimetic drug that binds to the calcium receptor on the parathyroid gland, fooling it into reducing PTH secretion in cases of hyperparathyroidism.

Knowledge Check Question

Question: In addition to treating hypercalcemia, what pathophysiology are bisphosphonates effective in preventing?

Answer: In addition to treating hypercalcemia, bisphosphonates reduce the pathologic risk of fractures by preventing osteoclastic bone resorption.

Case Connection

Thinking back to TG, how do you explain his bone pain?

You explain that TG's bone pain could indicate metastasis, but that might not be the case. Certain types of lung cancer can cause high calcium levels by secreting a hormone that affects bones and causes pain. "Your abdominal pain could also be related to the high calcium level. We can treat high calcium levels with medication to help prevent the bones from breaking down," you tell TG and his wife. You order a bone scan to look for metastases and refer TG for biopsy, staging, and treatment. Knowing that the couple will have many questions, you schedule a follow-up visit to review the new results in a few days and continue forming the plan to treat his pain.



Summary of Hypercalcemia

- Hypercalcemia is defined as a high serum calcium concentration (>10.3 mg/dL).
- Symptoms of hypercalcemia include neuropsychiatric, cardiovascular, gastrointestinal, renal, and musculoskeletal complaints.
- Common causes of hypercalcemia include malignancy, primary hyperparathyroidism, and granulomatous diseases (eg, sarcoidosis, tuberculosis).
- Malignancies may cause hypercalcemia by either secreting parathyroid hormone (PTH)-like proteins or by metastatic invasion of bone.
- Hypercalcemia raises the activation potential of cells and shortens cardiomyocyte repolarization time, leading to a short QT interval on the ECG.
- Measure serum calcium to establish the diagnosis, then do further testing to determine the underlying cause of hypercalcemia.
- Additional tests can include measuring levels of serum parathyroid hormone, vitamin D, calcitriol, phosphate, and parathyroid hormone-related protein (PTHrP).
- Treatment of chronic, asymptomatic hypercalcemia involves correcting the underlying issue and avoiding triggers such as thiazides, lithium, and dehydration.
- Treatment for acute, symptomatic hypercalcemia is based on severity and focuses on volume expansion with normal saline. The addition of bisphosphonates, calcitonin, and loop diuretics may also be indicated by the degree of hypercalcemia.

Check Your Knowledge

Part 2 Hypocalcemia and Hypercalcemia Review Questions

Question 1

A male presents to the emergency department complaining of tingling around his lips, fingers, and toes as well as twitching of his eyes. He receives an ECG that shows QT prolongation. Laboratory studies show a calcium concentration of 5.0 mg/dL. Which of the following is the most important next step in the management of this patient?

- A. Intravenous calcium gluconate
- B. Magnesium replenishment
- C. Oral calcium carbonate
- D. Potassium replenishment
- E. Vitamin D supplementation

Question 2

A female visits her primary care clinician two weeks after total thyroidectomy for papillary thyroid carcinoma. One of her complaints is that she seems to be experiencing “pins and needles” that won’t go away. After laboratory tests are ordered, the patient’s serum calcium is found to be 6.7 mg/dL. Which of the following responses might you expect after tapping the side of this patient’s face?

- A. Involuntary grunting noises



- B. Seizures
- C. Spasmodic contraction of the wrist and carpal muscles
- D. Tinnitus
- E. Twitching of the facial muscles

Question 3

Which of the following is not a common cause of hypocalcemia?

- A. Furosemide
- B. Hyperparathyroidism
- C. Hypomagnesemia
- D. Hypoparathyroidism
- E. Vitamin D deficiency

Question 4

Which of the following is the least common symptom of hypercalcemia?

- A. Coma
- B. Constipation
- C. Depression
- D. Kidney stones
- E. Tetany

Question 5

A 65-year-old male with shortness of breath and edema is diagnosed with congestive heart failure. He complains of constipation and lethargy and seems confused. Noted on his record is a serum calcium level of 11.2 mg/dL. What medication used for heart failure could potentially worsen this patient's presentation?

- A. Acetazolamide
- B. Calcitonin
- C. Chlorthalidone
- D. Digoxin
- E. Furosemide

Question 6

A 72-year-old male presents for a broken arm and his history reveals that he has had several other fractures in the past year. His serum calcium level is 10.7 mg/dL. The observed hypercalcemia would be most likely caused by metastasis of which of the following cancers?

- A. Colorectal adenocarcinoma
- B. Melanoma
- C. Multiple myeloma



- D. Pilocytic astrocytoma
- E. Small cell carcinoma of the lung

Question 7

A 32-year-old female comes to the clinic with complaints of fatigue, joint pain, incessant coughing, and various skin rashes on her chest and abdomen. Lab findings show that her serum calcium level is 12.0 mg/dL. A skin biopsy shows sarcoidosis. Which of the following drugs would best address her condition?

- A. Bisphosphonates
- B. Calcitonin
- C. Lithium
- D. Methotrexate
- E. Prednisone

Question 8

A 61-year-old male is brought to the emergency department with confusion, delirium, and stupor. His medical history is notable for a kidney transplant operation that occurred a year ago. His serum calcium level is 14.8 mg/dL. Which of the following is the best next step in management?

- A. Bisphosphonates
- B. Dialysis
- C. Lithium
- D. Normal saline
- E. Thiazide diuretics

Part 2 Hypocalcemia and Hypercalcemia Review Questions and Answers

Question 1

A male presents to the emergency department complaining of tingling around his lips, fingers, and toes as well as twitching of his eyes. He receives an ECG that shows QT prolongation. Laboratory studies show a calcium concentration of 5.0 mg/dL. Which of the following is the most important next step in the management of this patient?

- A. Intravenous calcium gluconate**
- B. Magnesium replenishment
- C. Oral calcium carbonate
- D. Potassium replenishment
- E. Vitamin D supplementation

Feedback: The correct answer is intravenous calcium gluconate (A). Acute symptomatic hypocalcemia requires immediate treatment with intravenous (IV) calcium gluconate to correct the acute abnormality and prevent cardiac arrhythmias, bronchospasms, and laryngospasms.



Magnesium replenishment (B) is useful for the long-term replenishment of calcium if hypomagnesemia is present, but the first step is IV calcium gluconate. Oral calcium carbonate (C) is useful for the long-term correction of hypocalcemia, but IV calcium gluconate is required in the acute setting. Potassium replenishment (D) is not indicated in the case of acute symptomatic hypocalcemia; also, this question does not deal with a potassium disturbance. Vitamin D supplementation (E) is useful to help correct hypocalcemia in the long-term, but not in the acute symptomatic setting.

Question 2

A female visits her primary care clinician two weeks after total thyroidectomy for papillary thyroid carcinoma. One of her complaints is that she seems to be experiencing “pins and needles” that won’t go away. After laboratory tests are ordered, the patient’s serum calcium is found to be 6.7 mg/dL. Which of the following responses might you expect after tapping the side of this patient’s face?

- A. Involuntary grunting noises
- B. Seizures
- C. Spasmodic contraction of the wrist and carpal muscles
- D. Tinnitus
- E. Twitching of the facial muscles**

Feedback: The correct answer is twitching of the facial muscles (E). A symptomatic hypocalcemic patient can present with Chvostek sign, which is twitching of the facial muscles when tapping the facial nerve. Involuntary grunting noises (A) are not related to hypocalcemia. Seizures (B) can be a symptom of hypocalcemia, but they are not elicited by tapping a patient’s face. Spasmodic contraction of the wrist and carpal muscles (C) describes the Trousseau sign and occurs when a blood pressure cuff is inflated on a patient’s arm. Tinnitus (D) is ringing of the ears, and it is not related to hypocalcemia or Chvostek sign.

Question 3

Which of the following is not a common cause of hypocalcemia?

- A. Furosemide
- B. Hyperparathyroidism**
- C. Hypomagnesemia
- D. Hypoparathyroidism
- E. Vitamin D deficiency

Feedback: The correct answer is hyperparathyroidism (B). Hyperparathyroidism would result in hypercalcemia, not hypocalcemia, whereas all the other choices can cause hypocalcemia. This is because parathyroid hormone (PTH) stimulates the release of calcium from the bones into the bloodstream and the reabsorption of calcium by the kidneys. Furosemide (A) is a loop diuretic that can cause hypocalcemia because it impairs renal calcium absorption. Hypomagnesemia (C) can cause hypocalcemia because magnesium is necessary for the proper secretion of PTH. Hypoparathyroidism (D) can cause hypocalcemia because PTH functions to increase serum calcium levels by acting on bones, kidneys, and the gastrointestinal tract. Vitamin D is necessary



for the reabsorption of calcium in the small intestine, and so vitamin D deficiency (E) can therefore cause hypocalcemia.

Question 4

Which of the following is the least common symptom of hypercalcemia?

- A. Coma
- B. Constipation
- C. Depression
- D. Kidney stones
- E. Tetany**

Feedback: The correct answer is tetany (E). Tetany is a classic symptom of hypocalcemia, not hypercalcemia. As levels of calcium decrease and levels of phosphate increase, spasms occurs, as seen in tetany. A classic mnemonic for consequences of hypercalcemia is stones, bones, moans, and groans. Coma (A) is a neurological symptom that may present in severe hypercalcemia. Constipation (B) is a common symptom, because excessive calcium levels will result in dehydration. Dehydration will decrease gastrointestinal motility, resulting in constipation. Depression (C) is a common psychiatric symptom that can be caused by hypercalcemia. Kidney stones (D) are a common symptom of hypercalcemia, as excess calcium may concentrate in the renal system and precipitate into stones.

Question 5

A 65-year-old male with shortness of breath and edema is diagnosed with congestive heart failure. He complains of constipation and lethargy and seems confused. Noted on his record is a serum calcium level of 11.2 mg/dL. What medication used for heart failure could potentially worsen this patient's presentation?

- A. Acetazolamide
- B. Calcitonin
- C. Chlorthalidone**
- D. Digoxin
- E. Furosemide

Feedback: The correct answer is chlorthalidone (C). Thiazides inhibit the sodium chloride cotransporter in the distal convoluted tubule. Thiazide diuretics may be used to help treat osteoporosis by increasing Ca^{2+} reabsorption. Therefore, thiazides may cause hypercalcemia. Acetazolamide (A) is a carbonic anhydrase inhibitor and mild diuretic. It does not cause hypercalcemia but may cause hypokalemia and hyperammonemia. Calcitonin (B) is used to decrease serum calcium levels and increase serum phosphate levels. Digoxin (D) inhibits the $\text{Na}^+/\text{K}^+-\text{ATPase}$ but has little effect on calcium concentrations. Furosemide (E) is a loop diuretic that increases, not decreases, renal excretion of calcium, so it is used to treat hypercalcemia.



Question 6

A 72-year-old male presents for a broken arm and his history reveals that he has had several other fractures in the past year. His serum calcium level is 10.7 mg/dL. The observed hypercalcemia would be most likely caused by metastasis of which of the following cancers?

- A. Colorectal adenocarcinoma
- B. Melanoma
- C. Multiple myeloma**
- D. Pilocytic astrocytoma
- E. Small cell carcinoma of the lung

Feedback: The correct answer is multiple myeloma (C). Multiple myeloma is the most common primary bone malignancy, in which there is a neoplastic proliferation of plasma cells producing a monoclonal immunoglobulin. Multiple myeloma causes hypercalcemia by extensive bone destruction. The myeloma cells locally express potent cytokines, which increase osteoclastic bone resorption and ultimately cause bone destruction. Colorectal adenocarcinoma (A) is not known to cause hypercalcemia. Melanoma (B) is the most malignant skin cancer and is not a cause of hypercalcemia. Pilocytic astrocytoma (D) is a benign pediatric brain tumor with a good prognosis and no association with hypercalcemia. Small cell carcinoma of the lung (E) is a tumor that can secrete excess antidiuretic hormone (ADH) and cause hyponatremia but does not generally cause hypercalcemia.

Question 7

A 32-year-old female comes to the clinic with complaints of fatigue, joint pain, incessant coughing, and various skin rashes on her chest and abdomen. Lab findings show that her serum calcium level is 12.0 mg/dL. A skin biopsy shows sarcoidosis. Which of the following drugs would best address her condition?

- A. Bisphosphonates
- B. Calcitonin
- C. Lithium
- D. Methotrexate
- E. Prednisone**

Feedback: The correct answer is prednisone (E). Sarcoidosis is an immune-mediated inflammatory disease that is characterized by noncaseating granulomas in multiple organs, especially in the lungs. Prednisone is a glucocorticoid that is used to treat sarcoidosis because of the drug's anti-inflammatory actions and its ability to decrease calcium levels. Glucocorticoids decrease calcitriol production from activated mononuclear cells. Bisphosphonates (A) and calcitonin (B) can be used to treat symptomatic hypercalcemia, but they do not address the underlying granulomatous disease (eg, sarcoidosis). Lithium (C) is used to treat bipolar disorder. Lithium causes hypercalcemia by an unknown mechanism. It is thought that lithium may directly stimulate PTH production. Methotrexate (D) is a folate antagonist that inhibits dihydrofolate reductase, leading to decreased formation of the folates necessary for DNA synthesis. Methotrexate depletes folates,



causing bone marrow toxicity. Small doses of folinic acid (leucovorin) are required to replenish the folate.

Question 8

A 61-year-old male is brought to the emergency department with confusion, delirium, and stupor. His medical history is notable for a kidney transplant operation that occurred a year ago. His serum calcium level is 14.8 mg/dL. Which of the following is the best next step in management?

- A. Bisphosphonates
- B. Dialysis
- C. Lithium
- D. Normal saline**
- E. Thiazide diuretics

Feedback: The correct answer is normal saline (D). Fluid administration is the most important step in treating hypercalcemia. Correction of acute symptomatic hypercalcemia should begin with correction of hypovolemia with 0.9% saline. Hypovolemia prevents normal calciuresis. Therefore, correcting volume depletion will increase renal calcium excretion. Bisphosphonates (A) are used to treat mild to moderate hypercalcemia. Bisphosphonates are used as an adjuvant therapy after normal saline correction. They are particularly effective in treating hypercalcemia caused by increased bone reabsorption, but they work more slowly. Dialysis (B) is rarely used for hypercalcemia. Lithium (C), a drug used to treat bipolar disorder, causes hypercalcemia by unknown mechanisms. Thiazide diuretics (E) are often used for patients with hypertension or congestive heart failure; they reduce renal calcium excretion and can cause hypercalcemia. Loop diuretics, in contrast, increase renal calcium excretion and can be used to treat hypercalcemia.



Function and Disorders of the Parathyroid Gland Part 3 of 4: Hypophosphatemia and Hyperphosphatemia

Overview

Some content and images for this lecture are adapted from ScholarRx and used with permission.

Learning Objectives

This page covers the following learning objectives:

Pharmacology

4. Detail the pharmacological interventions for managing hypophosphatemia, including the use of oral and intravenous phosphate preparations and FGF23 antibody Burosumab.
5. Describe the role of phosphate binders in managing serum phosphate levels with a focus on mechanism of action and adverse effects.

Hypophosphatemia

Case Connection

TL is a 39-year-old male who is brought to the emergency department after being found lying on the ground in the park. He smells of alcohol and is incoherent. His vitals include a temperature of 98.8°F, heart rate of 120 beats/min, and blood pressure of 150/92 mm Hg. On physical exam, TL appears malnourished. After you order bloodwork, you begin IV fluids with dextrose. You are texted by the lab with a critical value: TL's phosphate is low.

How will you explain TL's current status to him? Consider your answer as you read, and we'll revisit TL at the end of the brick.

What Is Hypophosphatemia?

Phosphate (PO_4^{3-}) is a crucial compound in the human body, affecting a broad spectrum of cellular processes. Just a few of its essential roles include regulating enzyme function in the form of adenosine triphosphate (ATP), forming the backbone of DNA and RNA nucleotides, controlling oxygen release from red blood cells as 2,3-diphosphoglyceric acid (2,3-DPG), and providing structural support in bone.



The normal range of serum phosphate values in adults is 3.0 to 4.5 mg/dL (the lower normal value varies somewhat by age). Hypophosphatemia is defined as a serum phosphate level lower than 3 mg/dL. Not surprisingly, severe decreases in serum phosphate can have widespread and dangerous consequences.

Knowledge Check Question

Question: What is the normal range for serum phosphate levels in adults?

Answer: The normal range for serum phosphate levels in adults is 3.0 to 4.5 mg/dL.

How Do Patients With Hypophosphatemia Present?

Hypophosphatemia is usually asymptomatic and picked up on lab testing, often in malnourished or hospitalized patients. It only becomes truly symptomatic once levels drop lower than 1.0 mg/dL (Figure 1).

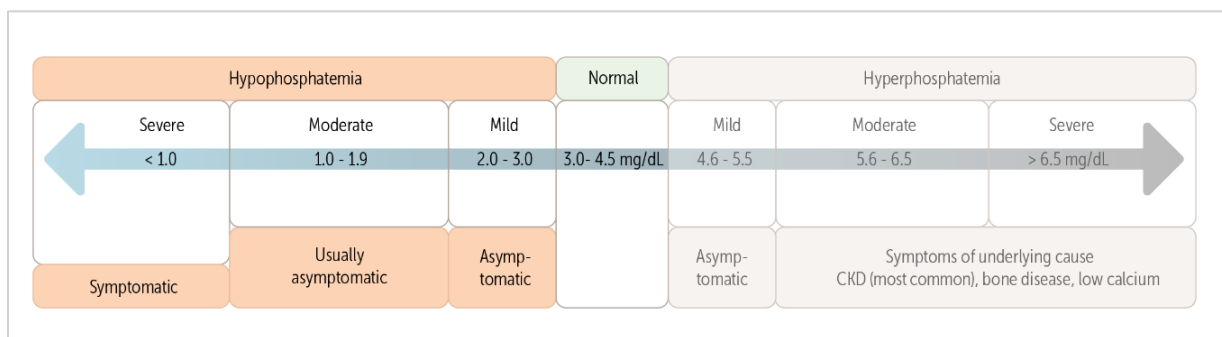


Figure 1 Credit: ©ScholarRx

In patients with severe prolonged hypophosphatemia, depletion of ATP may cause **muscle weakness** or dysphagia from reduced muscle contractility. Decreased cardiac contractility can cause **heart failure**, and decreased diaphragmatic function may cause respiratory failure. Less often, there are central nervous system (CNS) effects like confusion (encephalopathy), numbness, seizures, or coma. Severe hypophosphatemia can also cause rhabdomyolysis (muscle lysis). All these are uncommon effects mostly seen in severely ill hospitalized patients after starvation or metabolic disease.

Knowledge Check Question

Question: How do patients with hypophosphatemia typically present?

Answer: Mild hypophosphatemia typically is asymptomatic; symptoms generally are not evident clinically until the condition is severe.

What Is the Pathophysiology of Hypophosphatemia?

Hypophosphatemia is mostly seen in severely ill patients with sepsis, trauma, severe malnutrition, or chronic alcohol use. Milder cases can be seen in hormone abnormalities. The consequences can be severe, especially for critically ill patients.

Causes

There are three main reasons for an abnormally low level of phosphate in the blood:

- increased renal excretion
- insufficient dietary intake or gut absorption
- redistribution (shifting of phosphate stores from extracellular to intracellular space)

We'll look at each in turn. See below to learn more.

Increased Renal Phosphate Excretion

Phosphate excretion occurs primarily via the kidneys. Most phosphate that is filtered is reabsorbed in the proximal convoluted tubule (80%) or in the distal tubule (10%) by sodium-phosphate cotransporters. The rest (10%) is excreted into the urine. Excessive excretion of phosphate occurs when renal phosphate reabsorption is impaired.

The most common cause of decreased renal phosphate reabsorption is **osmotic diuresis**, as occurs in hyperglycemic patients with diabetes, where high filtration of glucose into the tubule pulls water and electrolytes with it into the urine, causing polyuria and electrolyte abnormalities like hypokalemia and hypophosphatemia.

Renal phosphate reabsorption is decreased by parathyroid hormone (PTH), so **primary hyperparathyroidism** may cause mild hypophosphatemia. (The more important effect is hypercalcemia.) The hypophosphatemia is mild because PTH also increases bone resorption of phosphate.

Last, some **medications**, such as the diuretic acetazolamide (a carbonic anhydrase inhibitor), may lead to hypophosphatemia because they inhibit the reabsorption of sodium and phosphate at the proximal convoluted tubule (Figure 2).

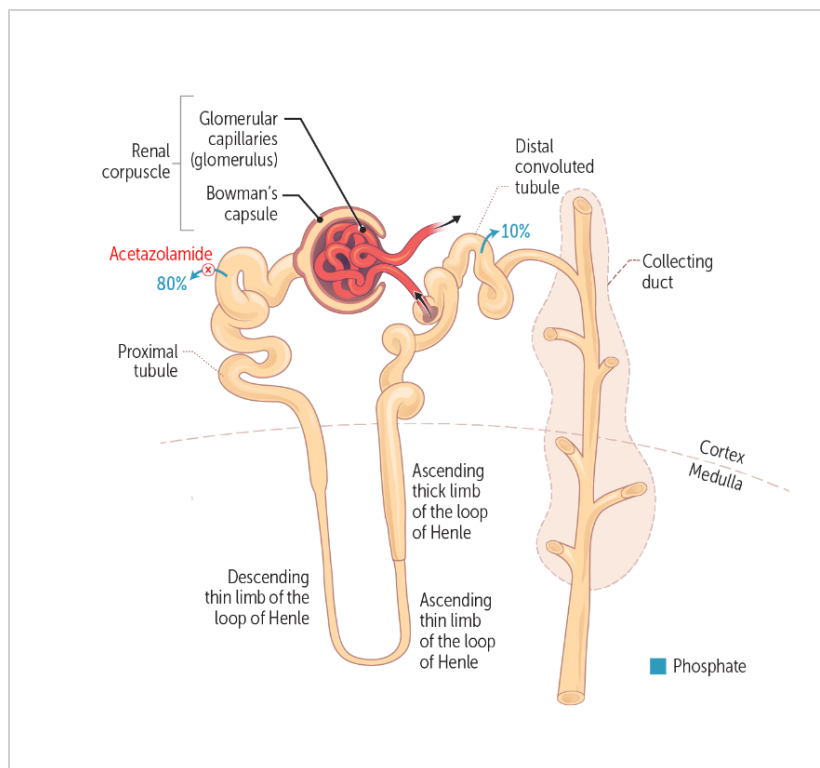


Figure 2 Credit: ©ScholarRx

Other drugs, such as tenofovir, can cause proximal tubular cell dysfunction, leading to hypophosphatemia from urinary phosphate wasting.

Clinical Correlation



Fanconi syndrome is a genetic disorder of the renal proximal tubule whereby various substances—including glucose, bicarbonate, potassium, and phosphate—are unable to be reabsorbed, causing their loss in the urine. It can lead to growth defects and bone disorders.

Decreased Phosphate Intake and Gut Absorption

The most common overall cause of hypophosphatemia is heavy alcohol use, which directly inhibits phosphate absorption in the gut. Heavy alcohol use is often coupled with poor dietary intake, which makes the hypophosphatemia worse. Patients admitted to emergency rooms with alcohol withdrawal and severe intoxication often have combined hypophosphatemia, hypokalemia, hypomagnesemia, and metabolic alkalosis and need extensive replacement of these electrolytes.

In the absence of alcohol use, dietary insufficiency is a relatively uncommon cause of hypophosphatemia in the United States. About 700 mg of phosphate is the required intake daily, but a normal diet provides about double that. If dietary phosphate intake is



insufficient, it is almost always associated with malnutrition secondary to eating disorders or socioeconomic limitations in access to food.

Less commonly, dietary phosphate consumption can be normal, but absorption in the gut can be impaired. This may be due to malabsorption syndromes (steatorrhea) or by aluminum-, calcium-, or magnesium-based antacids that bind phosphate and prevent its absorption. It may also be the result of reduced circulating vitamin D (rickets). Vitamin D stimulates phosphate and calcium absorption from the intestine, so vitamin D deficiency, usually seen in developing countries, can lead to hypophosphatemia by decreasing gut absorption of phosphate.

Knowledge Check Question

Question: What types of medications can impair gut phosphate absorption?

Answer: Antacids, specifically those that are aluminum-, calcium-, or magnesium-based.

Intracellular Redistribution of Phosphate

As with serum potassium levels, serum phosphate levels may fall rapidly if phosphate rapidly shifts from outside to inside the cells. This can occur during **insulin administration**, while treating diabetic ketoacidosis. But unlike potassium shifting, this is not a simple intracellular shifting done by activating sodium-potassium ATPase pumps. In phosphate's case, insulin administration stimulates glycolysis, which uses circulating phosphate to create phosphorylated intermediate metabolites. This may cause serum phosphate levels to fall, especially in those whose phosphate levels are low to begin with. This is why phosphate may need to be replaced during treatment of diabetic ketoacidosis.

Insulin levels may also rise during **refeeding of severely malnourished patients** (eg, persons with eating disorders and marasmus). During severe malnutrition, β -islet cells of the pancreas store synthesized insulin in vesicles, which accumulates without a hyperglycemic signal to release the vesicles. Upon refeeding such patients, the high carbohydrate intake stimulates massive insulin release and causes a fall in serum phosphate by shifting it into cells because of increased cellular demand.

A less common situation is **hungry bones syndrome**. After parathyroidectomy for hyperparathyroidism, there may be sudden depositing of calcium and phosphate into bones, leading to marked hypophosphatemia and hypocalcemia.



Physiologic Effects and Consequences

Hypophosphatemia can affect multiple systems. The effects of hypophosphatemia result from depleted levels of two key regulatory molecules that have phosphate: ATP and 2,3-diphosphoglyceric acid (2,3-DPG).

ATP Depletion

When ATP-dependent functions fail, many problems follow. Think muscle contraction, cardiopulmonary function, and nerve impulse transmission.

Lack of ATP causes **muscle weakness** by impairing contractility of both skeletal and smooth muscle. ATP depletion prevents the Na⁺/K⁺-ATPase and Ca²⁺-ATPase pumps from working effectively, which means they are no longer able to maintain muscle cell integrity. This leads to mitochondrial dysfunction, activation of proteases, and production of reactive oxygen species, all of which will contribute to the death of myocytes by rhabdomyolysis (muscle breakdown). This can be particularly important in the **respiratory muscles**, where weakness can cause respiratory failure in sick patients in the intensive care unit.

In the heart, impairment of Na⁺/K⁺-ATPase reduces cardiac myocyte contractility and conduction, leading to arrhythmias and heart failure.

ATP depletion in the central nervous system disrupts neurotransmission, leading to a range of mental state changes, from irritability to coma.

2,3-Diphosphoglycerate Depletion

Hypophosphatemia also decreases 2,3-diphosphoglycerate (2,3-DPG). This substance normally decreases hemoglobin's affinity for oxygen, facilitating oxygen delivery to the tissues. A decrease in 2,3-DPG increases hemoglobin's affinity for oxygen, shifting the oxyhemoglobin dissociation curve to the left and impairing the release of oxygen to tissues (Figure 3).

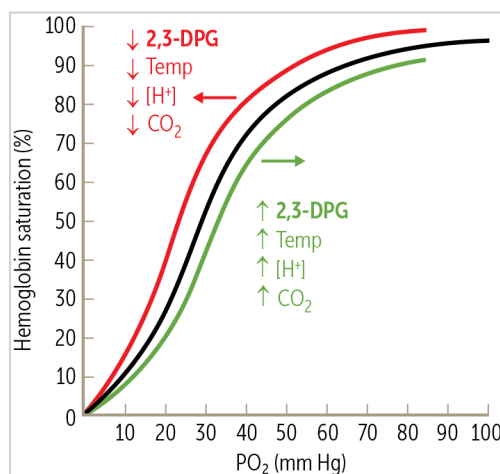


Figure 3 Credit: ©ScholarRx

This impaired oxygen release can create big problems in hospitalized patients with respiratory failure, particularly when combined with the respiratory muscle weakness that is also caused by hypophosphatemia.

Knowledge Check Question

Question: Why is hypophosphatemia a problem in patients with respiratory failure?

Answer: Limited release of oxygen to tissues because of 2,3-DPG depletion and respiratory muscle weakness

How Do We Diagnose Hypophosphatemia?

A combination of physical examination and lab findings are usually sufficient to make a diagnosis. Rarely, skeletal abnormalities, which can be detected on plain x-ray, may accompany hypophosphatemia.

Physical Examination

Patients with mild-to-moderate hypophosphatemia are diagnosed incidentally because they do not present with signs and symptoms. However, severe cases manifest with altered mentation, weakness, respiratory depression, and features of heart failure such as pulmonary edema and shock.

Laboratory Tests

A low serum phosphate value confirms hypophosphatemia, and it may be the only abnormality in mild cases. You may also detect evidence to support a clinical history of underlying cause, such as hyperglycemia and hyperketonemia in diabetic ketoacidosis.

A **spot urine phosphate measurement** or a 24-hour urine collection can sometimes be helpful to diagnose uncertain cases. If urine phosphate is low, think of decreased phosphate intake or decreased reabsorption of phosphate in the gut. If the urine phosphate is high, excessive renal loss is the cause.

Knowledge Check Question

Question: What does 24-hour urine phosphate tell you about the cause of hypophosphatemia?

Answer: A low level points to decreased phosphate intake or decreased phosphate absorption in the gut, whereas a high level points to increased renal loss.

How Do We Manage Hypophosphatemia?

Treatment of hypophosphatemia centers first on treating the underlying cause. For most patients, that treatment will soon correct phosphate levels. Phosphate repletion is reserved mainly for severe cases, as in diabetic ketoacidosis (DKA) with falling serum levels.

Oral phosphate supplements are preferred because they have a lower risk of overdose and hyperphosphatemia; nonetheless, IV supplements are indicated in symptomatic severe hypophosphatemia. However, giving too much phosphate can quickly lower serum calcium levels, causing new problems.

Knowledge Check Question

Question: Why are oral phosphate supplements preferred to IV supplements?

Answer: Oral phosphate supplements are associated with a lower risk of overdose and hyperphosphatemia than are IV supplements.

Case Connection

Thinking back to TL, how do you explain his current status to him?

You write orders for TL's phosphate repletion. After he becomes more coherent, you sit with him and explain that his phosphate levels are low because of his poor eating and excessive alcohol intake. You explain that this condition can be life threatening. TL is offered admission to an alcohol rehabilitation program. He says he knows it will be difficult, but he's willing to give it a shot.

Summary of Hypophosphatemia

- Hypophosphatemia is defined as a serum phosphate level less than 3 mg/dL.
- Mild and moderate hypophosphatemia are normally asymptomatic.
- In severe hypophosphatemia, symptoms include fatigue and muscle weakness and, less commonly, cardiac arrhythmias, respiratory distress, and altered mental status.
- The most common cause of decreased gastrointestinal absorption is chronic alcohol use (often paired with poor dietary intake).
- The most common cause of excessive renal excretion is osmotic diuresis, as seen in hyperglycemia from diabetic ketoacidosis.



- Most clinical effects of hypophosphatemia come from the depletion of adenosine triphosphate (ATP) or 2,3-diphosphoglyceric acid (2,3-DPG), leading to cellular energy depletion and the inability of red blood cells to release oxygen to the tissues.
- The cause of hypophosphatemia is usually evident from the patient's history.
- In uncertain cases, the 24-hour urinary phosphate level can determine if the etiology is renal phosphate wasting or reduced gut absorption.
- Hypophosphatemia is usually treated with oral phosphate supplements, but intravenous (IV) phosphate may be used in symptomatic patients.

Hyperphosphatemia

Case Connection

"Why do I have to take calcium? I thought only women took calcium pills. Don't I already take enough pills?" JH says during his follow-up visit at the renal clinic. You are seeing him as he prepares to initiate hemodialysis. JH has chronic kidney disease secondary to diabetes mellitus. On JH's most recent set of routine labs, his creatinine and phosphate levels are flagged as being high.

How will you explain to JH why his phosphate is high and why he needs to take calcium? Consider your answers as you read, and we'll revisit JH at the end of the brick.

What Is Hyperphosphatemia?

Phosphate (PO_4^{3-}) is a crucial compound in the human body, affecting a broad spectrum of cellular processes. Too much, however, as happens in patients with **chronic kidney disease (CKD)**, can be a bad thing. It reduces the serum calcium level and causes a secondary increase in parathyroid hormone, leading to bone disease.

Why is hyperphosphatemia seen in patients with chronic kidney disease? It happens because the elimination of phosphate is surprisingly simple: it is excreted only by the kidneys. This little fact is extremely relevant when discussing hyperphosphatemia, which is defined as a serum phosphate level above the normal range of 3.0 to 4.5 mg/dL. So, as we talk about the causes, effects, and management of hyperphosphatemia, keep your eye on the kidney!



Knowledge Check Question

Question: What is the level of serum phosphate in hyperphosphatemia?

Answer: The level of serum phosphate in hyperphosphatemia is any level above 4.5 mg/dL.

How Do Patients With Hyperphosphatemia Present?

Hyperphosphatemia on its own is typically asymptomatic, but it is often accompanied by symptoms of the disease that causes it (Figure 1).

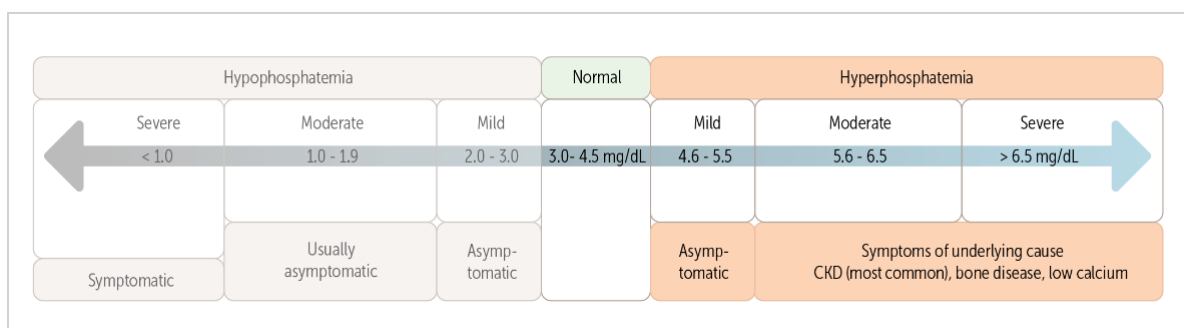


Figure 1 Credit: ©ScholarRx

In patients with hyperphosphatemia secondary to CKD, patients may present with other symptoms of CKD, such as edema. If the CKD is far advanced, they may show symptoms of uremia: fatigue, shortness of breath, anorexia, nausea, and vomiting.

Untreated hyperphosphatemia can contribute to a disease called **renal osteodystrophy** (osteo = bone, dystrophy = degeneration), resulting in complaints of bone pain, fractures, or painful calcium-phosphate deposits in the skin. These patients can also develop calcifications within their vasculature.

Hyperphosphatemia may lead to low serum calcium (hypocalcemia), causing symptoms such as hyperirritability, tremors, and seizures. Patients with hyperphosphatemia are also at risk for developing kidney stones.

Knowledge Check Question

Question: Patients with hyperphosphatemia often present with symptoms of which disease?

Answer: Patients with hyperphosphatemia often present with symptoms of CKD, because hyperphosphatemia is often secondary to this disease



What Is the Pathophysiology of Hyperphosphatemia?

In this section, we will first look at the causes of hyperphosphatemia, which are largely related to reduced excretion, increased intake, or redistribution of phosphate (shift from intracellular to extracellular space). Then we will review the main consequences of hyperphosphatemia, which are not from the excess serum phosphate, but rather, its lowering of the serum calcium.

Causes

We'll focus mostly on the hyperphosphatemia seen in patients with CKD. But there are three main mechanisms that can lead to hyperphosphatemia:

- **Chronic kidney disease**, causing failure of the kidney to clear phosphate
- Phosphate overload from the sudden death of cells, as seen in tumor cells lysing after chemotherapy or in muscle cell lysis
- Excess dietary intake or gut absorption (least common mechanism)

Reduced Renal Excretion of Phosphate

The kidneys play the key role in phosphate balance since they are the only organs that excrete phosphate. Loss of more than 50% to 75% of renal function, as is seen in **advanced CKD**, results in impaired phosphate excretion. As chronic kidney disease progresses and the glomerular filtration rate (GFR) declines, phosphate excretion falls. Below a certain threshold, the kidney is unable to excrete sufficient phosphate to maintain homeostasis, and hyperphosphatemia results. The risk of elevated serum phosphate in patients with CKD is higher if they consume a phosphate-rich diet (eg, meats, nuts, dairy products).

Another condition that lowers renal phosphate excretion is **hypoparathyroidism**. Low levels of parathyroid hormone (PTH) increase phosphate reabsorption in the kidney and may sometimes lead to mild hyperphosphatemia.

Knowledge Check Question

Question: What causes hyperphosphatemia in chronic kidney disease?

Answer: Hyperphosphatemia is caused by impaired excretion of phosphate from the kidneys.

Redistribution of Phosphate

An acute cause of hyperphosphatemia is an acute increase in serum phosphate after the sudden lysis of cells, with the intracellular phosphate spilling into the blood. This happens



in rhabdomyolysis (muscle cell lysis after trauma or stress) and tumor lysis syndrome (lysis of tumor cells after chemotherapy). In turn, both of these conditions can cause acute kidney injury, which reduces renal excretion of phosphate and thus leads to hyperphosphatemia.

Increased Gut Absorption of Phosphate

Phosphate is present in many foods and beverages, including meats, dairy products, nuts, and colas (not most noncola drinks). While eating too much phosphate alone will not cause hyperphosphatemia, CKD patients with impaired phosphate excretion will fare worse if **consuming** too much phosphate. These patients generally have to follow low-phosphate diets.

Similarly, vitamin D increases both phosphate and calcium absorption from the gut and may result in hyperphosphatemia if ingested in toxic quantities. Patients with kidney disease who use phosphorous-containing enemas or laxatives are also at high risk of hyperphosphatemia.

Physiologic Effects and Consequences

High serum phosphate directly lowers serum calcium by **binding and precipitating calcium in the peripheral tissues**. Hypocalcemia, in turn, causes an elevation in serum PTH, a hormone designed to help the body maintain bone and calcium homeostasis. (This PTH elevation is known as secondary hyperparathyroidism.) The increase in PTH causes an increase in calcium resorption from the bony matrix, which, in turn, reduces the structural integrity of the bony skeleton and thus leads to **renal osteodystrophy**.

In CKD, the serum calcium is also decreased because the conversion of 25-OH to 1,25-OH vitamin D (calcitriol) in the kidneys is reduced. So treatment of renal osteodystrophy is directed at simultaneously lowering the serum phosphate and increasing the levels of calcitriol. These mechanisms are summarized in Figure 2.

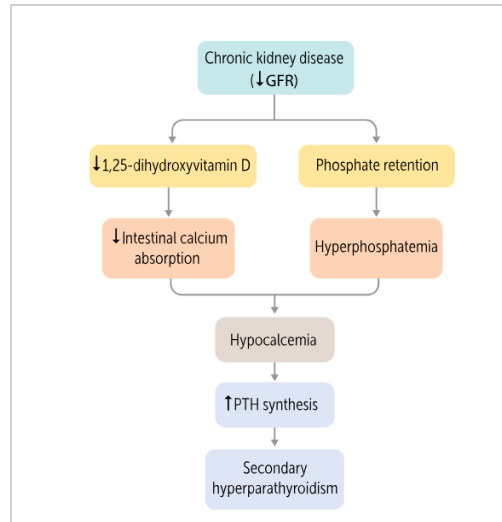


Figure 2. Image credit: ScholarRx

Eventually, if the calcium levels remain low, the parathyroid gland may hypertrophy and become resistant to regulation, constantly churning out PTH. This is termed tertiary hyperparathyroidism and usually requires surgical removal of the parathyroid glands.

How Do We Diagnose Hyperphosphatemia?

We rely on a thorough physical examination, laboratory findings, and a skeletal survey to make a diagnosis of hyperphosphatemia and its underlying causes.

Physical Examination

The physical exam findings seen in hyperphosphatemia are those of the underlying causes, most commonly CKD. Patients might have risk factors for CKD such as longstanding, poorly controlled hypertension and diabetes mellitus. Evidence of uremia, such as altered mentation, volume overload, and anemia, may be present. Features of hypocalcemia, such as perioral numbness or tingling, arrhythmias, tremors, and seizures, may also be present. ECG is required in those with hypocalcemia for early detection of fatal arrhythmia.

Laboratory Tests

Since this is initially an asymptomatic disease, hyperphosphatemia is diagnosed by routine serum chemistry. A full metabolic panel and PTH must be ordered in all patients with suspected hyperphosphatemia.

Evidence of CKD includes elevated BUN and creatinine and, sometimes, hyperkalemia on the metabolic panel. Serum uric acid and potassium levels will be elevated in those with tumor lysis syndrome. It is imperative to check parathyroid hormone levels in those with hypocalcemia. High serum phosphate and low serum calcium with elevated PTH suggest



secondary hyperparathyroidism in CKD patients. The blood levels of all of these substances will guide medical management.

Skeletal Survey

X-ray of the bones may show subperiosteal lytic bone lesions called brown tumors or osteitis fibrosa cystica. These lesions suggest high bone turnover in CKD patients, although these tumors occur very late in the disease. Soft tissue calcifications can also be seen on radiography.

Knowledge Check Question

Question: Serum levels of which substances are elevated in tumor lysis syndrome?

Answer: Serum levels of phosphate, uric acid, and potassium are elevated in tumor lysis syndrome.

How Do We Manage Hyperphosphatemia?

As with all electrolyte disorders, the treatment must address the underlying disease process. In patients with CKD, it's critical to reduce the serum phosphate level in order to prevent the complications of renal osteodystrophy.

Diet

Patients with CKD are advised to reduce their intake of foods rich in phosphate, such as processed meats, dairy products, chocolate, beer, nuts, and colas.

Medication

If dietary restriction proves inadequate, patients will also be given gut **phosphate binders**. These dietary supplements bind phosphate in the lumen of the gut, preventing phosphate absorption. The bound phosphate is then excreted in the stool. Examples include calcium carbonate or calcium acetate, as well as noncalcium phosphate binders (sevelamer, lanthanum). The latter are more expensive but may be preferable, because giving calcium to a patient with high phosphate levels may cause the precipitation of calcium phosphate in tissues.

For patients with secondary hyperparathyroidism, other treatments may include calcitriol or cinacalcet to increase serum calcium and reduce PTH secretion.

Knowledge Check Question

Question: How do phosphate binders help treat hyperphosphatemia in CKD?



Answer: They reduce dietary phosphate absorption in the lumen of the gut.

Dialysis

Peritoneal dialysis and hemodialysis can remove excess phosphate from circulation, but they do so inefficiently. Dialysis is an inefficient treatment of hyperphosphatemia because so much phosphate is stored in the bone; this means it's not a substitute for diet and binders.

Hyperphosphatemia alone is not an indication for dialysis, which is instead usually started because of additional findings of end-stage renal disease (ESRD), including severe fluid overload, hyperkalemia, acidosis, or uremic symptoms.

Case Connection

Thinking back to JH, how do you explain why his phosphate is high and why he needs to take calcium?

You explain to JH that his kidneys are functioning so poorly that they are failing to excrete phosphate, so it is building up in his blood. Over time, this can lead to bone problems. Calcium binds the phosphate in the gut so that it can be excreted in the stool. "Well, I suppose that makes sense," he says. "It's kind of like the phosphate is having to take a detour to get out of the body?" You think that this is a good time to go over JH's medication list and explain why he is taking each medication.

Summary of Hyperphosphatemia

- Hyperphosphatemia is defined as a serum phosphate concentration >4.5 mg/dL.
- Hyperphosphatemia usually doesn't cause symptoms itself, but it is often accompanied by symptoms of the underlying disease process.
- Soft tissue calcifications may be present in patients with chronic kidney disease.
- Symptoms of hypocalcemia may be evident in patients with co-occurring hypocalcemia.
- Hyperphosphatemia is mostly due to reduced renal excretion and can be worsened by high phosphate intake or lysis of cells, the latter of which causes movement of intracellular phosphate into the serum.
- Hyperphosphatemia in chronic kidney disease is caused by reduced phosphate excretion by the kidneys and can lead to hypocalcemia, secondary hyperparathyroidism, and bone disease.
- Hyperphosphatemia is diagnosed by finding a high serum phosphate concentration.



- Physical exam and laboratory findings in hyperphosphatemia are results of the underlying causes, such as chronic kidney disease (CKD), and complications, such as hypocalcemia and osteodystrophy.
- Treatment of chronic hyperphosphatemia in chronic kidney disease consists primarily of a low-phosphate diet and oral phosphate binders.

Check Your Knowledge

Part 3 Hypophosphatemia and Hyperphosphatemia Review Questions

Question 1

A 63-year-old male presents to the emergency room after several weeks of palpitations. He has had generalized anorexia, fatigue, and malaise for several months. He has a long history of chronic heartburn, for which he takes over-the-counter antacids several times per day. Labs are significant for serum phosphate of 1.6 mg/dL and a normal serum calcium level. His physical exam is normal. Which of the following is the most likely cause of his hypophosphatemia?

- A. Decreased intestinal phosphate absorption
- B. Decreased phosphate intake
- C. Decreased synthesis of phosphate
- D. Increased renal phosphate excretion
- E. Reduced bone release of phosphate

Question 2

Which of the following would be least likely in a phosphate-depleted patient?

- A. Decreased oxygen delivery to tissues
- B. Hypoparathyroidism
- C. Hypoventilation
- D. Muscle necrosis
- E. Muscle weakness

Question 3

A 68-year-old male with type 1 diabetes mellitus comes to the emergency department with fatigue and weakness. His glucose is 430 mg/dL, potassium 3.1 mmol/L, and phosphate 0.9 mg/dL. Ketones are detected in his blood and urine. What caused the electrolyte imbalances in this patient and led to his current condition?

- A. Decreased intestinal phosphate absorption
- B. Decreased phosphate production from internal stores
- C. Decreased vitamin D levels



- D. Increased renal phosphate excretion
- E. Movement of phosphate into cells

Question 4

A 59-year-old female with a history of diabetes mellitus comes to the emergency room with dizziness and abdominal pain. Her muscle and neurologic exams are normal. The electrocardiogram (ECG) is normal. She is found to have a serum glucose of 450 mg/dL and blood pH of 7.28 with positive serum ketones and is treated for diabetic ketoacidosis with insulin and saline. During treatment, she is noted to have a serum phosphate level of 2.3 mg/dL. Which of the following is the most appropriate next step in management of her hypophosphatemia?

- A. Check 24-hour urine phosphate.
- B. Give additional insulin.
- C. Give intravenous phosphate.
- D. Give oral phosphate.
- E. No additional management steps are needed.

Question 5

A 33-year-old schoolteacher with a history of relapsed Hodgkin lymphoma presents to the emergency department with a 12-hour history of vomiting, new-onset seizures, and an acute decline in mental status. His first treatment with cisplatin chemotherapy was 24 hours ago. Lab studies show a serum phosphate of 6.9 mg/dL, elevated from 3.5 mg/dL in labs drawn prior to yesterday's infusion. The serum creatinine is 1.2 mg/dL. What is the likely source of this patient's new-onset hyperphosphatemia?

- A. Chronic kidney disease
- B. Extensive use of phosphate supplements
- C. Increased gut phosphate reabsorption
- D. Lysis of malignant cells
- E. Noncompliance with dietary phosphate restrictions

Question 6

A 78-year-old male with a history of chronic kidney disease who is now undergoing hemodialysis complains of pain in his back, wrists, and hips. He has a history of chronic hyperphosphatemia with poor adherence to his low-phosphate diet. Which of the following serum findings is most likely in this patient?

- A. High calcitriol
- B. High ionized calcium
- C. High parathyroid hormone



- D. Low potassium
- E. Normal total calcium levels with low ionized calcium

Question 7

Which of the following foods is safest to eat for a patient with severe chronic kidney disease and persistent hyperphosphatemia?

- A. Almonds
- B. Ginger ale
- C. Peanut butter
- D. Salami
- E. Yogurt

Question 8

A 42-year-old female comes to the emergency department with fatigue and confusion. She has a long history of type 2 diabetes mellitus and was noted to have an increased serum creatinine and 4 g of proteinuria/24 hours last year. She is disoriented to time and place. Her exam now shows bibasilar crackles and severe bipedal edema. Serum studies show a phosphate of 12.2 mg/dL, mild hypocalcemia and hyperkalemia, a blood urea nitrogen (BUN) of 115 mg/dL, and a serum creatinine (Cr) level of 8.2 mg/dL. Which of the following is the best next step in treatment?

- A. Dietary phosphate restriction
- B. Hemodialysis
- C. Normal saline infusion
- D. Oral calcium supplementation
- E. Oral phosphate binders

Part 3 Hypophosphatemia and Hyperphosphatemia Questions and Answers

Question 1

A 63-year-old male presents to the emergency room after several weeks of palpitations. He has had generalized anorexia, fatigue, and malaise for several months. He has a long history of chronic heartburn, for which he takes over-the-counter antacids several times per day. Labs are significant for serum phosphate of 1.6 mg/dL and a normal serum calcium level. His physical exam is normal. Which of the following is the most likely cause of his hypophosphatemia?

- A. Decreased intestinal phosphate absorption
- B. Decreased phosphate intake



- C. Decreased synthesis of phosphate
- D. Increased renal phosphate excretion
- E. Reduced bone release of phosphate

Feedback: The correct answer is decreased intestinal phosphate absorption (A). The patient took over-the-counter antacids that likely contained aluminum, calcium, or magnesium compounds. These decrease the absorption of phosphate. This would not be an issue for healthy patients who eat excess phosphate as part of a balanced diet, but this patient likely had reduced intake due to anorexia. The trigger, however, was excess antacid use. Phosphate is not synthesized (C) per se, but instead stored in bone and released as needed to maintain constant blood levels. There is no history to suggest the malnutrition associated with hypophosphatemia due to decreased intake (B). Increased renal phosphate excretion (D) is primarily seen in hyperparathyroidism and genetic conditions affecting the renal tubules, such as Fanconi syndrome. Bone release of phosphate (E) happens with parathyroid hormone, but there is no evidence for hypoparathyroidism here.

Question 2

Which of the following would be least likely in a phosphate-depleted patient?

- A. Decreased oxygen delivery to tissues
- B. Hypoparathyroidism**
- C. Hypoventilation
- D. Muscle necrosis
- E. Muscle weakness

Feedback: The correct answer is hypoparathyroidism (B). Hyperparathyroidism is a cause of hypophosphatemia, but the phosphate level does not directly suppress the parathyroid gland. All the other answers are highly likely. Low phosphate reduces 2,3-diphosphoglyceric acid, decreasing oxygen release from red blood cells (A); weakens respiratory muscles, causing hypoventilation (C); causes rhabdomyolysis, or muscle necrosis (D); and commonly causes muscle weakness via adenosine triphosphate depletion (E).

Question 3

A 68-year-old male with type 1 diabetes mellitus comes to the emergency department with fatigue and weakness. His glucose is 430 mg/dL, potassium 3.1 mmol/L, and phosphate 0.9 mg/dL. Ketones are detected in his blood and urine. What caused the electrolyte imbalances in this patient and led to his current condition?

- A. Decreased intestinal phosphate absorption
- B. Decreased phosphate production from internal stores



- C. Decreased vitamin D levels
- D. Increased renal phosphate excretion**
- E. Movement of phosphate into cells

Feedback: The correct answer is increased renal phosphate excretion (D). This patient is experiencing diabetic ketoacidosis (DKA), marked by severe hyperglycemia and ketoacids in the blood. The kidney undergoes osmotic diuresis, dumping glucose along with potassium, phosphate, sodium, and water into the urine. This leads to severe body depletion of all of these substances, and hypophosphatemia is a common result. In DKA, intestinal phosphate absorption is generally not affected (A), and mobilization of calcium and phosphate from bone stores is likewise not a contributing factor (B). Insulin used to treat DKA will drive phosphate into cells (E), but this patient has not yet been treated. Vitamin D deficiency (C) would decrease gut phosphate absorption if present, but there is no reason for levels to be low here.

Question 4

A 59-year-old female with a history of diabetes mellitus comes to the emergency room with dizziness and abdominal pain. Her muscle and neurologic exams are normal. The electrocardiogram (ECG) is normal. She is found to have a serum glucose of 450 mg/dL and blood pH of 7.28 with positive serum ketones and is treated for diabetic ketoacidosis with insulin and saline. During treatment, she is noted to have a serum phosphate level of 2.3 mg/dL. Which of the following is the most appropriate next step in management of her hypophosphatemia?

- A. Check 24-hour urine phosphate.
- B. Give additional insulin.
- C. Give intravenous phosphate.
- D. Give oral phosphate.**
- E. No additional management steps are needed.

Feedback: The correct answer is give oral phosphate (D). The patient's phosphate has dropped to the slightly subnormal range as part of treating her DKA, so it needs repletion. Given the normal ECG and lack of symptoms, oral therapy is fine and no intravenous treatment is needed (C). Insulin is needed for DKA treatment but not for phosphate treatment (B). A 24-hour urine phosphate test would not be needed, as her loss is via osmotic diuresis during DKA (A). Low phosphate levels should be treated, as they may drop more during DKA therapy (E).

Question 5

A 33-year-old schoolteacher with a history of relapsed Hodgkin lymphoma presents to the emergency department with a 12-hour history of vomiting, new-onset seizures, and



an acute decline in mental status. His first treatment with cisplatin chemotherapy was 24 hours ago. Lab studies show a serum phosphate of 6.9 mg/dL, elevated from 3.5 mg/dL in labs drawn prior to yesterday's infusion. The serum creatinine is 1.2 mg/dL. What is the likely source of this patient's new-onset hyperphosphatemia?

- A. Chronic kidney disease
- B. Extensive use of phosphate supplements
- C. Increased gut phosphate reabsorption
- D. Lysis of malignant cells**
- E. Noncompliance with dietary phosphate restrictions

Feedback: The correct answer is massive lysis of malignant cells (D). This vignette describes the classic presentation of tumor lysis syndrome: a sudden change in mental status accompanied with massive electrolyte disturbances in a patient starting induction chemotherapy for a hematologic malignancy. Recall that phosphate is the major intracellular anion and that rapid and massive cell death triggers the increase of serum phosphate. The normal serum creatinine makes chronic kidney disease unlikely (A). Phosphate overuse (B) and high phosphate intake in food (E) are rare causes of hyperphosphatemia unless there is concurrent CKD. High gut phosphate absorption is uncommon (C).

Question 6

A 78-year-old male with a history of chronic kidney disease who is now undergoing hemodialysis complains of pain in his back, wrists, and hips. He has a history of chronic hyperphosphatemia with poor adherence to his low-phosphate diet. Which of the following serum findings is most likely in this patient?

- A. High calcitriol
- B. High ionized calcium
- C. High parathyroid hormone**
- D. Low potassium
- E. Normal total calcium levels with low ionized calcium

Feedback: The correct answer is high parathyroid hormone level (C). The high phosphate level will lower the ionized and total calcium levels (B and E), which will stimulate the secretion of PTH. Patients with CKD have low conversion of 25- to 1,25-dihydroxy vitamin D (calcitriol), so the calcitriol level is low, not high (A). Patients with CKD have high, not low, serum potassium (D).

Question 7

Which of the following foods is safest to eat for a patient with severe chronic kidney disease and persistent hyperphosphatemia?



- A. Almonds
- B. Ginger ale**
- C. Peanut butter
- D. Salami
- E. Yogurt

Feedback: The correct answer is ginger ale (B). Ginger ale is free of phosphorus and so would be the best choice. Foods and beverages high in phosphorus are not recommended for patients with CKD who have impaired phosphate excretion. Products higher in phosphorous include colas (but not most noncola drinks); nuts, including almonds (A) and peanuts (C); processed and organ meats, like salami (D); and most dairy products, including yogurt (E).

Question 8

A 42-year-old female comes to the emergency department with fatigue and confusion. She has a long history of type 2 diabetes mellitus and was noted to have an increased serum creatinine and 4 g of proteinuria/24 hours last year. She is disoriented to time and place. Her exam now shows bibasilar crackles and severe bipedal edema. Serum studies show a phosphate of 12.2 mg/dL, mild hypocalcemia and hyperkalemia, a blood urea nitrogen (BUN) of 115 mg/dL, and a serum creatinine (Cr) level of 8.2 mg/dL. Which of the following is the best next step in treatment?

- A. Dietary phosphate restriction
- B. Hemodialysis**
- C. Normal saline infusion
- D. Oral calcium supplementation
- E. Oral phosphate binders

Feedback: The correct answer is hemodialysis (B). This patient likely has chronic kidney disease because of diabetic nephropathy, and now she has signs of uremia (confusion, hyperphosphatemia, elevated BUN and creatinine). In patients with hyperphosphatemia with uremic symptoms (edema, confusion), dialysis is the preferred therapy. A low-phosphate diet (A), binding phosphate in the gut via oral calcium supplementation (D), and oral phosphate binders (E) will be appropriate after initial dialysis stabilizes the patient or for patients without symptoms. Normal saline (C) would be hazardous in a fluid overloaded (edematous) patient with limited renal function.



Function and Disorders of the Parathyroid Gland Part 4 of 4: Hypoparathyroidism and Hyperparathyroidism

Overview

Some content and images for this lecture are adapted from ScholarRx and used with permission.

Learning Objectives

This page covers the following learning objectives:

Pathology

1. Discuss the etiopathogenesis of hyperparathyroidism.
2. Explain the pathogenesis of brown tumor in the setting of chronic renal failure.
4. Describe the morphological features noted in i) Parathyroid adenoma, ii) Brown tumor. Differentiate parathyroid hyperplasia and carcinoma.

Clinical Medicine

1. Discuss the clinical presentations of hyperparathyroidism and hypoparathyroidism.
4. Discuss investigative studies that are required for further evaluation of both hyperparathyroidism and hypoparathyroidism.

Pharmacology

6. Explain the use of PTH analogs and calcium/vitamin D supplementation in the management of hypoparathyroidism.
7. Discuss the pharmacological management of hyperparathyroidism, including the roles of bisphosphonates, cinacalcet and denosumab.

Hypoparathyroidism

Case Connection

EF, a 62-year-old male with chronic alcohol overuse, is brought to your office by his daughter. “He’s not eating well at all and he is much weaker than when I last saw him several months ago,” she says. “And, yes, he is still drinking.” On exam, EF is severely underweight and when you take his blood pressure, you note that the muscles of his hand involuntarily contract. In addition, he has diffuse, symmetric upper and lower extremity muscle weakness. On review of his labs, you notice that his calcium level is 7.7 mg/dL (normal: 8.4-10.2 mg/dL) and serum phosphate level is elevated; his ionized calcium level is low.

What are the best next tests to determine the etiology of the EF’s low calcium?
How would you treat EF? Consider your answers as you read, and we’ll revisit EF at the end of the brick.

What Is Hypoparathyroidism?

Hypoparathyroidism is a low concentration of parathyroid hormone (PTH) in the blood. PTH’s main function is to regulate calcium in the body. When PTH is reduced, serum calcium levels fall, leading to the signs and symptoms of hypocalcemia. Low PTH most frequently results from removal of the parathyroid glands during thyroid surgery, but there are other causes related to drug treatment and diseases.

How Do Patients With Hypoparathyroidism Present?

The symptoms of hypoparathyroidism are all caused by hypocalcemia. Interestingly, the level of symptoms that a patient presents with can vary from person to person and depends on the degree and duration of the hypocalcemia. Acute hypocalcemia may cause symptoms with smaller reductions in serum calcium than would chronic hypocalcemia. This is because the body is not given time to acclimate.

Postoperative (acute) hypoparathyroidism

Postoperative (acute) hypoparathyroidism can present with acute hypocalcemia, which needs to be managed urgently. This often presents with tetany (neuromuscular irritability) leading to paresthesias, muscle cramps, fatigue, laryngospasms (wheezing), or even seizures. Mood symptoms—including hyperirritability, anxiety, or mood swings—are also possible.

Chronic hypoparathyroidism

Chronic hypoparathyroidism, in contrast, can present with milder versions of the acute symptoms listed above along with other symptoms indicative of longstanding hypocalcemia and low PTH. This may include movement disorders (parkinsonism, hemiballismus) caused by calcifications in the basal ganglia and decreased vision caused by cataracts.

Knowledge Check Question

Question: What is the main symptom of acute hypoparathyroidism?

Answer: Tetany is the main symptom of acute hypoparathyroidism.

What Causes Hypoparathyroidism?

Hypoparathyroidism is caused by an absent or disordered parathyroid gland (Figure 1). In hypoparathyroidism, the production, secretion, and activity of PTH is impaired.

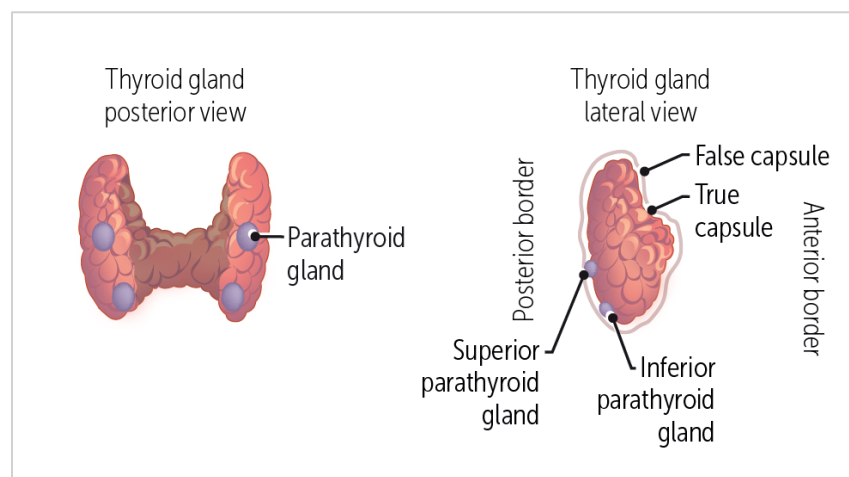


Figure 1 Credit: ©ScholarRx

So how does an absent or poorly functioning parathyroid gland occur?

- **Surgery and related treatments:** As we've noted, the most common cause is removal or damage to the parathyroid glands during neck surgery, such as the resection of a thyroid tumor or another head or neck malignancy. Damage may also occur during radiation exposure, or, rarely, during radioiodine treatment for hyperthyroidism.
- **Autoimmune disease:** Besides iatrogenic hypoparathyroidism, the most common cause is autoimmune disease, usually presenting in childhood or adolescence. It results from mutations in the autoimmune regulator (AIRE) gene.



- **Hypomagnesemia:** A more transient form of low PTH occurs in hypomagnesemia, seen often in malnourished patients or those with chronic alcohol overuse. Magnesium is needed for the parathyroid glands to synthesize and release PTH, so hypomagnesemia may cause hypoparathyroidism and hypocalcemia. Magnesium replacement corrects the problem.

A related condition is **pseudohypoparathyroidism**. This is a rare genetic defect in PTH receptors that leads to tissues resisting PTH. This causes some of the same problems of hypoparathyroidism, including hypocalcemia and hyperphosphatemia, but with a high PTH level. Children with this condition may be shorter and may have shorter bones in their hands and feet.

Knowledge Check Question

Question: What is the most common cause of hypoparathyroidism?

Answer: Neck surgery is the most common cause of hypoparathyroidism.

How Do We Diagnose Hypoparathyroidism?

Patients with hypoparathyroidism may have physical exam signs of hypocalcemia. Further lab testing will then determine whether they have hypoparathyroidism.

Physical Examination

Neuromuscular irritability is an important sign of hypocalcemia, which can occur secondary to hypoparathyroidism. Two tests that can be used to clinically diagnose hypocalcemia on physical exam are the Chvostek sign and the Trousseau sign:

- Chvostek sign is the twitching of facial muscles in response to tapping on the facial nerve.
- Trousseau sign is a carpopedal spasm triggered by inflating a blood pressure cuff to a pressure above the systolic blood pressure for 3 minutes.

Laboratory Findings

Patients With Suspected Hypocalcemia. The obvious starting point is to order a serum calcium. The normal range is 8.4 to 10.2 mg/dL. Because albumin binds calcium, patients who have low serum albumin (eg, liver failure or nephrotic syndrome) should have a free (ionized) calcium checked to confirm that a low total calcium actually reflects true hypocalcemia. The ionized calcium will always be low in true hypocalcemia.



Patients Found to Have Hypocalcemia. A further workup will determine whether the hypocalcemia is secondary to hypoparathyroidism or another disorder. A further workup may include:

- **Serum albumin** (or an ionized calcium level) to ensure the free calcium level is low
- **Serum parathyroid hormone** level to document a low value indicating hypoparathyroidism
- **Serum phosphate level**, because hypoparathyroidism will usually cause mild hyperphosphatemia
- **Serum magnesium**, because hypomagnesemia causes hypocalcemia
- **Serum 25-OH vitamin D**, because vitamin D deficiency causes hypocalcemia

Knowledge Check Question

Question: Name a mineral deficiency—other than calcium—and a vitamin deficiency that may cause hypocalcemia.

Answer: Magnesium and vitamin D deficiencies may cause hypocalcemia.

How Do We Manage Hypoparathyroidism?

As we've explained, hypoparathyroidism can present acutely or chronically, and treatment depends on which form shows up.

Acute hypoparathyroidism (postoperative) frequently causes tetany or neuromuscular findings that need to be addressed immediately, and it often requires intravenous (IV) calcium administration.

For **chronic hypoparathyroidism**, first-line treatment is **calcium supplementation** with oral calcium carbonate or calcium citrate. Vitamin D, specifically in its active form **calcitriol**, is also an important piece of treatment, as it helps with calcium absorption. Thiazide diuretics may be of use in treatment of hypoparathyroidism because they decrease urinary excretion of calcium and enhance reabsorption.

Knowledge Check Question

Question: What type of diuretic may be used as part of the treatment for hypoparathyroidism?

Answer: Thiazide diuretics may be used as part of the treatment for hypoparathyroidism.

Case Connection

Thinking back to EF, what are the best next tests to determine the etiology of his low calcium? How would you treat EF?

"It's probably the magnesium," you say to the student who is rotating through your office. "Magnesium? But his problem is his calcium, not magnesium," she responds. You explain that although it is true that EF's symptoms and signs are caused by hypocalcemia, a low magnesium level, resulting from his poor nutrition and alcohol use, is the root of the problem. You further explain that a low serum magnesium impairs the synthesis and release of PTH and, in turn, a low PTH causes low serum calcium and elevated phosphate. Follow-up laboratory values confirm hypomagnesemia, hyperphosphatemia, and low PTH. "Let's start by giving him supplemental magnesium," you say. With magnesium supplementation, EF's magnesium normalizes, as does his serum calcium. His symptoms improve. "Now we need to address his alcohol use," the student says. "Precisely," you respond.

Summary of Hypoparathyroidism

- Hypoparathyroidism is an abnormally low concentration of parathyroid hormone (PTH) in the blood; this leads to hypocalcemia.
- Hypoparathyroidism most commonly presents with symptoms of hypocalcemia (eg, muscle weakness and muscle spasms).
- The most common etiology of hypoparathyroidism is incidental damage during neck surgery, especially thyroid surgery, but other causes include autoimmune disease or radiation.
- Patients with hypoparathyroidism have hypocalcemia, hyperphosphatemia, and low serum PTH levels.
- First-line treatment of hypoparathyroidism includes calcium and vitamin D supplementation.

Hyperparathyroidism

Case Connection

CE is a 57-year-old female with well-controlled hypothyroidism who is having her second episode of kidney stones this year. She had been following a kidney stone-prevention diet and drinking as much water as she could, but now she is asking you for relief from the pain. “My bones hurt constantly, and I am just so tired,” she says, “I can’t sleep at night. It’s really starting to get me down.” Her calcium level is elevated. An elevated PTH confirms hyperparathyroidism.

What will you tell CE about her condition and possible treatment? Consider your answer as you read, and we’ll revisit CE at the end of the brick.

Introduction

The parathyroid glands, usually four small glands located behind the thyroid gland in the neck, play a vital role in regulating calcium levels in the body. Their primary function is to produce and secrete parathyroid hormone (PTH), which helps maintain calcium homeostasis by acting on the bones, kidneys, and intestines.

Under normal circumstances, when blood calcium levels drop, the parathyroid glands sense this decrease and respond by increasing the production and secretion of PTH. PTH acts on the bones to release calcium into the bloodstream, enhances calcium reabsorption in the kidneys, and stimulates the production of active vitamin D, which promotes calcium absorption in the intestines.

What Is Hyperparathyroidism?

Hyperparathyroidism is a condition characterized by excessive secretion of PTH. It can be classified into primary, secondary, and tertiary types, each with distinct underlying causes and pathophysiology.

Knowledge Check Question

Question: What is the main stimulus for the parathyroid gland to secrete PTH?

Answer: Hypocalcemia is the main stimulus for PTH secretion.



Primary Hyperparathyroidism

Primary hyperparathyroidism occurs due to autonomous overproduction of PTH, usually from a single adenoma (benign tumor) or hyperplasia (enlargement) of one or more parathyroid glands. The majority of cases (85-95%) are caused by parathyroid adenomas, which are typically solitary and may vary in size. Less commonly, primary hyperplasia or, rarely, parathyroid carcinoma (1% of cases) may be responsible. Molecular defects, such as cyclin D1 gene inversions and MEN1 mutations, have been implicated in the development of sporadic adenomas.

Secondary Hyperparathyroidism

In most endocrine disorders, we think of secondary disease as too much of a stimulating hormone causing the target gland to oversecrete its hormone. For example, in secondary hyperthyroidism, excess TSH secreted by the anterior pituitary gland causes the thyroid to secrete too much thyroid hormone.

But that is not what we mean in secondary hyperparathyroidism—there is no stimulating hormone in excess. Instead, secondary hyperparathyroidism arises as a compensatory response to prolonged hypocalcemia, often secondary to conditions such as chronic kidney disease (renal failure). There is an appropriate rise in PTH, as parathyroid tries to correct hypocalcemia caused by; for example, chronic renal failure.

In chronic renal failure, impaired kidney function leads to decreased calcium excretion and reduced activation of vitamin D, resulting in hypocalcemia and increased PTH secretion. Secondary hyperparathyroidism may also occur in other conditions associated with calcium malabsorption or deficiency, such as malabsorption syndromes or vitamin D deficiency.

Tertiary Hyperparathyroidism

Tertiary hyperparathyroidism represents a persistent hypersecretion of PTH even after correction of the underlying cause of prolonged hypocalcemia, such as successful renal transplantation in patients with chronic kidney disease (long-standing secondary hyperparathyroidism). This results in refractory and autonomous production of PTH leading to hypercalcemia. Parts of the parathyroid gland synthesize PTH independently of serum calcium levels.

Knowledge Check Question

Question: What is secondary hyperparathyroidism?

Answer: In secondary hyperparathyroidism, a nonparathyroid disease such as CKD lowers the serum calcium, stimulating the parathyroid gland to secrete more PTH.

How Do Patients With Hyperparathyroidism Present?

The clinical manifestations of hyperparathyroidism vary but may include symptoms related to hypercalcemia, such as fatigue, weakness, bone pain, kidney stones, and neuropsychiatric symptoms. Additionally, untreated hyperparathyroidism can lead to long-term complications such as osteoporosis, kidney stones, and renal dysfunction.

Primary

Most patients with primary hyperparathyroidism have no symptoms. Symptomatic patients have clinical features of hypercalcemia. These include general weakness, CVS (LVH, Hypertension, shortened QT interval), nephrolithiasis, abdominal pain, bone pain, muscle pain, constipation, and psychological symptoms.

Symptoms Caused by Hypercalcemia. Symptomatic patients may get the familiar hypercalcemic findings of “bones, stones, groans, and moans” (Figure 1).

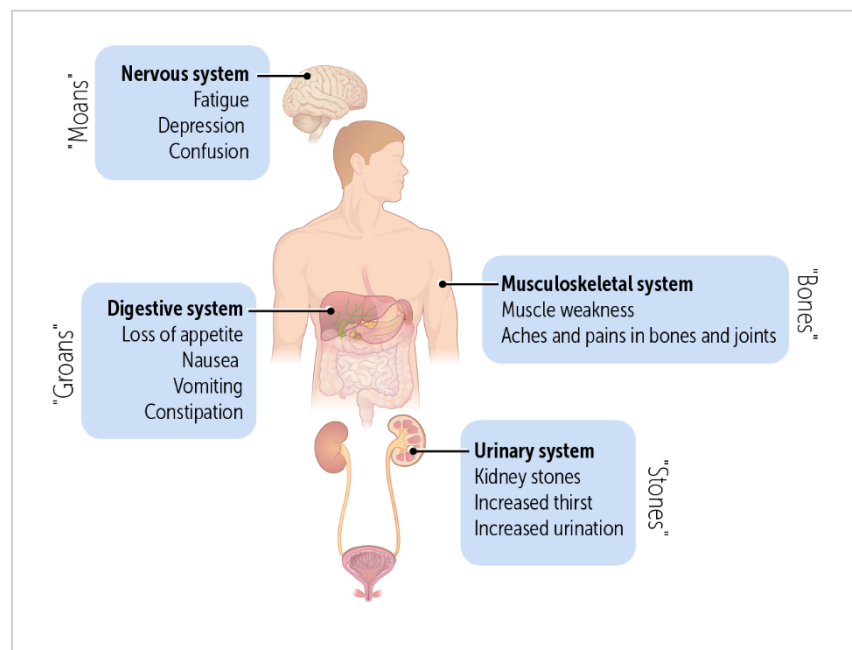


Figure 1 Credit: ©ScholarRx

This means they may develop:

- Bone pain and fractures
- Kidney stones (flank pain, hematuria)
- Constipation and cramps
- Confusion, fatigue, irritability, depression, memory problems

Additional worrisome complications of hypercalcemia include electrocardiogram changes (eg, shortened QT interval) and cardiac arrhythmias.

Knowledge Check Question

Question: Key signs of hyperparathyroidism-induced hypercalcemia: Painful bones, renal stones, abdominal groans, and psychiatric moans.

Answer: What are the signs of hyperparathyroidism-induced hypercalcemia?

Symptoms Caused by Bone Disease. Chronic excess PTH leads to **osteitis fibrosa cystica** (OFC), also known as von Recklinghausen disease of bone. Chronic high PTH levels cause osteoblasts to increase production of RANK-L (receptor activator of nuclear factor κ - β ligand). RANK-L on osteoblasts will then bind to RANK receptors on osteoclasts; the osteoclasts, now having been activated, will break down bone.

Chronic bone reabsorption may lead to brown cystic bone lesions, known as **brown tumors** (Figure 2).



Figure 2 Credit: Image courtesy of Khaoula BA et al. Case Rep Orthop 2011;2011:521578

This brown color comes from hemorrhages into the cystic space, causing hemosiderin to be deposited into the fibrous stroma.

Secondary and Tertiary

Most of the clinical signs and symptoms of secondary and tertiary hyperparathyroidism are caused by effects of excess PTH on bone in the setting of CKD or ESRD, as described above. Patients present with **bone and joint pain**, bone deformities and fractures, and decreased mobility.



Renal osteodystrophy is the collection of bone defects that are caused by CKD. The physiology of renal osteodystrophy is caused by a combination of high levels of PTH, acidosis, increased renal phosphate retention, and decreased renal production of calcitriol. The chronic high PTH levels cause high bone turnover and loss of bone calcium, leading to osteitis fibrosa cystica and osteomalacia. The acidosis dissolves bone hydroxyapatite. As hydroxyapatite is lost, bone matrix is demineralized, worsening the osteomalacia.

Knowledge Check Question

Question: How do most patients with secondary and tertiary hyperparathyroidism present?

Answer: Most present with bone and joint pain, deformities and fractures, and decreased mobility secondary to the effects of excess PTH on the bone.

In summary, hyperparathyroidism encompasses a spectrum of disorders characterized by dysregulation of PTH secretion and calcium homeostasis. Understanding the underlying causes and clinical implications of each type of hyperparathyroidism is essential for accurate diagnosis and management.

What Causes Hyperparathyroidism?

The cause of high PTH secretion varies depending on the subtype of hyperparathyroidism.

Primary

The most common cause of primary hyperparathyroidism is a **parathyroid gland adenoma**, a form of benign tumor. Rarer causes of primary hyperparathyroidism include parathyroid hyperplasia and parathyroid carcinoma.

Clinical Correlation



Parathyroid adenomas and hyperplasia are seen in multiple endocrine neoplasia (MEN) type 1 (MEN1) and MEN2A syndromes. MEN1 and MEN2A syndromes are caused by mutations in the tumor suppressor *MEN1* gene and the oncogene *RET*, respectively. Both syndromes can cause parathyroid adenoma or hyperplasia, resulting in primary hyperparathyroidism.



Secondary

As stated earlier, secondary hyperparathyroidism is caused by chronic hypocalcemia. Any cause of chronic hypocalcemia can be the culprit, but the most common cause is CKD. In CKD, the kidney stops making calcitriol (1,25 vitamin D), and there is high serum phosphate resulting from low glomerular phosphate filtration. Both factors lead to a drop in the serum calcium level, which then drives the high PTH levels. The increased PTH will subsequently return the serum calcium level back towards normal, so the patient will have low to normal calcium with high PTH.

Occasionally, secondary hyperparathyroidism may be seen in patients without CKD. This hypocalcemia may be caused by gut malabsorption or by vitamin D deficiency from a poor diet or lack of sun exposure.

Tertiary

Tertiary hyperparathyroidism is nearly always seen in patients with **end-stage renal disease (ESRD)** who receive dialysis therapy. Among these patients, those who receive transplants have more normal calcium homeostasis, so they do not usually develop tertiary hyperparathyroidism.

Knowledge Check Question

Question: What is the main cause of secondary hyperparathyroidism?

Answer: Chronic kidney disease is the main cause of secondary hyperparathyroidism.

Hypercalcemia of Malignancy

Hypercalcemia from malignancy is usually rapidly progressive; thus, rapidly rising calcium levels should increase suspicion of malignancy. Breast, lung, and kidney cancers should be considered, as should multiple myeloma, lymphoma, and leukemia. Excessive secretion of PTHrP is the most common cause of hypercalcemia of malignancy.

How Do We Diagnose Hyperparathyroidism?

The lab work is the key piece to diagnosing and differentiating primary, secondary, and tertiary hyperparathyroidism. Blood tests are used to measure PTH, calcium, and other minerals.

Laboratory Findings

Serum PTH levels in the blood will be high in all of these conditions, but the other lab values vary depending on the type.



Primary Hyperparathyroidism

Abnormal labs in primary hyperparathyroidism typically include increased PTH levels and hypercalcemia. Serum phosphate is either decreased or in the lower limit of the normal range. High PTH increases the levels of activated vitamin D (calcitriol). Urinary calcium, which is usually measured from a 24-hour urine collection, is increased because of high serum calcium. Urinary phosphate level is also increased because of PTH-mediated excretion of phosphate.

Secondary Hyperparathyroidism

Laboratory findings for secondary hyperparathyroidism include low-to-normal serum calcium and high phosphate and PTH levels. Calcitriol (1,25 dihydroxy vitamin D) levels are often low in those with CKD.

Tertiary Hyperparathyroidism

The lab findings for this form are caused by both ESRD and high PTH secretion. Patients often have hypercalcemia caused by parathyroid hyperplasia, with elevated serum phosphate levels caused by renal phosphate retention in ESRD.

Let's summarize by comparing what happens with calcium, phosphorus, PTH, and calcitriol in the different types of hyperparathyroidism (Table 1).

	Primary	Secondary	Tertiary
PTH	↑	↑	↑
Calcium	↑	↓ to normal	Usually ↑
Phosphate	↓	↑	Usually ↑
Calcitriol (1,25- dihydroxy vitamin D)	↑	↓	Usually ↓

Knowledge Check Question

Question: Besides PTH, what lab findings are elevated in tertiary hyperparathyroidism secondary to ESRD?

Answer: Serum phosphate and calcium are elevated in tertiary hyperparathyroidism secondary to ESRD.



Imaging

Imaging can help sort out the causes of hyperparathyroidism. On a neck ultrasound, a parathyroid adenoma (primary hyperparathyroidism) will appear as an oval-shaped, well-defined homogenous and hypoechoic mass, usually in just one of the four glands. Hyperplasia of all four glands (secondary and tertiary hyperparathyroidism) would appear as diffuse enlargement of the glands.

Another useful test for hyperparathyroidism is the technetium-99 sestamibi scan. The technetium-99 sestamibi radiotracer is taken up by parathyroid mitochondria. This scan is very good for identifying a singular parathyroid adenoma, though it is less accurate for parathyroid hyperplasia.

How Do We Manage Hyperparathyroidism?

Treatment depends on the type of hyperparathyroidism diagnosed.

Primary

Treating primary hyperparathyroidism requires addressing the underlying cause. If the primary hyperparathyroidism is caused by an adenoma, **surgical removal** is the definitive treatment. The surgery is curative but is technically difficult and sometimes fails because some overactive glands can remain. A common consequence of the surgery is hypoparathyroidism.

If medical management is selected instead of or in addition to surgery, **bisphosphonates** such as alendronate are indicated. They directly inactivate osteoclasts, causing a decrease in the amount of bone loss. Another option is **cinacalcet**, a calcimimetic drug that “fools” the parathyroid gland into turning off PTH secretion by binding to its calcium receptors.

Secondary

Treating patients with secondary hyperparathyroidism and CKD requires addressing the primary cause, which is chronic hypocalcemia. **Synthetic calcitriol** increases the calcitriol level in the blood, and the restriction of dietary phosphate and addition of **gut phosphate binders** lower the serum phosphate level. Both agents will increase the serum calcium level and remove the stimulus for high PTH secretion.

Tertiary

Treatment for tertiary hyperparathyroidism usually involves **surgical parathyroidectomy**. Removal of the parathyroid glands will address the autonomous hypersecretion of PTH that is refractory to medical treatment. Dialysis will relieve the acidosis that contributes to renal osteodystrophy. In some patients, the autonomous PTH secretion will subside after a kidney transplant.



Clinical Correlation



After parathyroidectomy, the bones will rapidly take up and mineralize calcium and other minerals. Patients need to be monitored for the “hungry bones syndrome,” which may show up as hypocalcemia, hypomagnesemia, hypophosphatemia, and hyperkalemia.

Knowledge Check Question

Question: What treatment is indicated for primary hyperparathyroidism?

Answer: Treating primary hyperparathyroidism includes surgical resection and bisphosphonates.

Case Connection

Thinking back to CE, what do you tell her about her condition and possible treatment?

You tell CE that her elevated PTH is most likely caused by an overgrowth of one or more of the tiny parathyroid glands in her neck that regulate calcium. You assure her that the symptoms she is experiencing are caused by the high calcium levels and that removal of the parathyroid glands will resolve that problem. You provide analgesics and instructions for straining her urine for stones. You also arrange a consultation with an endocrine surgeon for further evaluation and possible surgery.

Summary of Hyperparathyroidism

- Hyperparathyroidism is the abnormally high concentration of parathyroid hormone (PTH) in the blood; it can be primary, secondary, or tertiary.
- Primary hyperparathyroidism results from a parathyroid adenoma and causes hypercalcemia; patients are often asymptomatic, but may have constipation, renal stones, or confusion if the hypercalcemia is severe.
- Secondary hyperparathyroidism arises when parathyroid glands are chronically stimulated to release PTH in response to hypocalcemia caused by another condition, usually chronic kidney disease; patients may have low to normal serum calcium and bone pain caused by the chronic high PTH levels (renal osteodystrophy).



- Tertiary hyperparathyroidism occurs when long-standing secondary hyperparathyroidism progresses into autonomous hypersecretion of PTH despite hypercalcemia; patients may have bone pain caused by the chronic high PTH levels (renal osteodystrophy).
- Diagnostic testing includes serum PTH, calcium, and phosphate.
- Imaging includes neck ultrasound and sestamibi scanning.
- Treatment of primary hyperparathyroidism requires addressing the underlying cause, such as surgical removal of the adenoma or medical management.
- Secondary hyperparathyroidism is treated with synthetic calcitriol, restriction of dietary phosphate, and gut phosphate binders to correct hyperphosphatemia.
- Treatment for tertiary hyperparathyroidism involves parathyroidectomy.



Check Your Knowledge

Part 4 Hypoparathyroidism and Hyperparathyroidism Review Questions

Question 1

Which of the following is the most likely sign or symptom of primary hypoparathyroidism?

- A. Chvostek sign
- B. High parathyroid hormone
- C. Hypophosphatemia
- D. Increased urination
- E. Retinal disease

Question 2

What is the most common cause of hypoparathyroidism?

- A. Autoimmune disease
- B. Congenital problems
- C. Incidental damage
- D. Malignancy
- E. Radiation

Question 3

What is the first-line treatment for hypoparathyroidism?

- A. Calcium and vitamin D supplementation
- B. Parathyroidectomy
- C. Thiazide diuretics
- D. Thyroidectomy

Question 4

Which of the following is a cause of primary hyperparathyroidism?

- A. Chronic kidney disease
- B. DiGeorge syndrome
- C. Hypomagnesemia
- D. Parathyroid gland adenoma



Question 5

A patient comes to the clinic with complaints of heart palpitations. As he describes his palpitations, he also mentions that he has been experiencing fatigue, depression, frequent large volume urination, excessive thirst, and musculoskeletal pain. These symptoms reflect what electrolyte imbalance?

- A. Hypercalcemia
- B. Hypermagnesemia
- C. Hyponatremia
- D. Hypocalcemia
- E. Hypokalemia

Question 6

Which of the following is the most common cause of secondary hyperparathyroidism?

- A. Chronic kidney disease
- B. Pancoast tumor
- C. Parathyroid gland adenoma
- D. Parathyroid gland hyperplasia
- E. Thymoma

Question 7

A 48-year-old female comes to the clinic with depression, weakness, and fatigue. She has had two bone fractures within the past year. Her medical history is significant for chronic kidney disease, and she now receives hemodialysis three times a week. Her serum parathyroid hormone level is increased, and her serum calcium level is decreased. If untreated, which of the following is the most likely consequence of her disease?

- A. Autonomous secretion of parathyroid hormone by the parathyroid glands
- B. Hyperphosphatemia
- C. Hypocalcemia
- D. Parathyroid adenoma formation
- E. Parathyroid atrophy

Question 8

Which of the following is the most likely sign or symptom of hyperparathyroidism?

- A. Bone pain
- B. Dry skin
- C. Muscle cramping
- D. Paresthesias



E. Tetany

Part 4 Hypoparathyroidism and Hyperparathyroidism Review Questions and Answers

Question 1

Which of the following is the most likely sign or symptom of primary hypoparathyroidism?

- A. Chvostek sign
- B. High parathyroid hormone
- C. Hypophosphatemia
- D. Increased urination
- E. Retinal disease

Feedback: The correct answer is Chvostek sign (A). Chvostek sign is a physical exam finding in which facial muscles twitch in response to the facial nerve being tapped. This is a sign found in the setting of hypocalcemia. High parathyroid hormone (B) and hypophosphatemia (C) are incorrect because hypoparathyroidism would lead to low PTH and high serum phosphate. Increased urination (D) is incorrect because hyperparathyroidism, not hypoparathyroidism, leads to hypercalcemia and polyuria caused by nephrogenic diabetes insipidus. Retinal disease (E) is incorrect. Hypoparathyroidism leads to cataracts, which is a disease of the ocular lens, not the retina.

Question 2

What is the most common cause of hypoparathyroidism?

- A. Autoimmune disease
- B. Congenital problems
- C. Incidental damage
- D. Malignancy
- E. Radiation

Feedback: The correct answer is incidental damage (C). Incidental damage usually occurs in the setting of neck surgery, especially thyroid surgery. Autoimmune disease (A) is a cause of hypoparathyroidism, but it is less common. Congenital problems (B) are not a common cause of hypoparathyroidism. Malignancy (D) may induce hypoparathyroidism secondary to hypercalcemia of malignancy; this is not a common cause of hypoparathyroidism. Radiation (E) is another less common cause of hypoparathyroidism.



Question 3

What is the first-line treatment for hypoparathyroidism?

- A. Calcium and vitamin D supplementation
- B. Parathyroidectomy
- C. Thiazide diuretics
- D. Thyroidectomy

Feedback: The correct answer is calcium and vitamin D supplementation (A). These are intended to correct the hypocalcemia that is caused by hypoparathyroidism. Parathyroidectomy (B) would not improve hypoparathyroidism because it removes the glands that produce parathyroid hormone. Thiazide diuretics (C) may be used in conjunction with the calcium and vitamin D supplementation, but not as treatment alone. Thyroidectomy (D) would not help with hypoparathyroidism. Incidental damage to the parathyroids, usually secondary to thyroidectomy, is the most common cause of hypoparathyroidism.

Question 4

Which of the following is a cause of primary hyperparathyroidism?

- A. Chronic kidney disease
- B. DiGeorge syndrome
- C. Hypomagnesemia
- D. Parathyroid gland adenoma

Feedback: The correct answer is calcium and vitamin D supplementation (A). These are intended to correct the hypocalcemia that is caused by hypoparathyroidism. Parathyroidectomy (B) would not improve hypoparathyroidism because it removes the glands that produce parathyroid hormone. Thiazide diuretics (C) may be used in conjunction with the calcium and vitamin D supplementation, but not as treatment alone. Thyroidectomy (D) would not help with hypoparathyroidism. Incidental damage to the parathyroids, usually secondary to thyroidectomy, is the most common cause of hypoparathyroidism.

Question 5

A patient comes to the clinic with complaints of heart palpitations. As he describes his palpitations, he also mentions that he has been experiencing fatigue, depression, frequent large volume urination, excessive thirst, and musculoskeletal pain. These symptoms reflect what electrolyte imbalance?

- A. Hypercalcemia
- B. Hypermagnesemia



- C. Hypernatremia
- D. Hypocalcemia
- E. Hypokalemia

Feedback: The correct answer is hypercalcemia (A). In hyperparathyroidism, excess PTH causes increased calcium levels because of enhanced bone resorption by osteoclasts. Symptoms of hypercalcemia often include “painful bones, renal stones, abdominal groans, and psychiatric moans” and cause nephrogenic diabetes insipidus, leading to large urine volumes and thirst. Hypermagnesemia (B) has few symptoms, but if present they include weakness and absent tendon reflexes. Hypernatremia (C) is often seen because of unreplaced water that is lost from vomiting, diarrhea, and excessive sweating. Symptoms of hypernatremia include confusion and excessive thirst. Hypocalcemia (D) leads to paresthesia, tetany, and muscle cramping. Hypokalemia (E) leads to fatigue, myalgias, constipation, and cramps. Severe hypokalemia can cause arrhythmias and cardiac arrest.

Question 6

Which of the following is the most common cause of secondary hyperparathyroidism?

- A. Chronic kidney disease
- B. Pancoast tumor
- C. Parathyroid gland adenoma
- D. Parathyroid gland hyperplasia
- E. Thymoma

Feedback: The correct answer is chronic kidney disease (A). Secondary hyperparathyroidism occurs in CKD because of decreased phosphate excretion and loss of 1α -hydroxylase, which converts 25-hydroxy vitamin D to 1,25-dihydroxy vitamin D (the active form). Decreased 1,25-dihydroxy vitamin D results in hypocalcemia, further contributing to secondary hyperparathyroidism. Pancoast tumor (B) is not a cause of secondary hyperparathyroidism. Pancoast tumor is a tumor located in the apex of the lung that causes compression of the cervical sympathetic plexus, leading to Horner syndrome with a triad of ptosis, miosis, and anhidrosis on the ipsilateral side of the face. Parathyroid gland adenoma (C) is not a cause of secondary hyperparathyroidism; an adenoma of these glands would cause primary hyperparathyroidism. Parathyroid gland hyperplasia (D) is not a cause of secondary hyperparathyroidism; hyperplasia of these glands would cause primary hyperparathyroidism with excessive PTH release. Thymoma (E) is not a cause of secondary hyperparathyroidism. Thymomas are cancers made of epithelial cells of the thymus, with symptoms that include coughing, chest pain, dysphagia, and potentially voice hoarseness.



Question 7

A 48-year-old female comes to the clinic with depression, weakness, and fatigue. She has had two bone fractures within the past year. Her medical history is significant for chronic kidney disease, and she now receives hemodialysis three times a week. Her serum parathyroid hormone level is increased, and her serum calcium level is decreased. If untreated, which of the following is the most likely consequence of her disease?

- A. **Autonomous secretion of parathyroid hormone by the parathyroid glands**
- B. Hyperphosphatemia
- C. Hypocalcemia
- D. Parathyroid adenoma formation
- E. Parathyroid atrophy

Feedback: The correct answer is autonomous secretion of parathyroid hormone by the parathyroid glands (A). This describes tertiary hyperparathyroidism, which develops after long secondary hyperparathyroidism. The glands become unresponsive to serum calcium and secrete PTH autonomously. This most often happens in patients with severe CKD, especially in those on dialysis. Hyperphosphatemia (B) is a cause of secondary hyperparathyroidism and is not a consequence of excess PTH secretion, nor is hypocalcemia (C), so these are incorrect. Parathyroid adenoma formation (D) is a cause of primary, not tertiary hyperparathyroidism. Parathyroid atrophy (E) is a rare cause of hypoparathyroidism.

Question 8

Which of the following is the most likely sign or symptom of hyperparathyroidism?

- A. **Bone pain**
- B. Dry skin
- C. Muscle cramping
- D. Paresthesias
- E. Tetany

Feedback: The correct answer is bone pain (A). Symptoms of hyperparathyroidism include bone pain (from the high PTH causing bone turnover), confusion, abdominal pain, kidney stones, polyuria, constipation, and palpitations (from hypercalcemia). The remaining options are not symptoms of high PTH or of hypercalcemia. Dry skin (B) is a symptom of hypothyroidism caused by idiopathically decreased eccrine gland secretion. Muscle cramping (C) is a sign of hypokalemia, not hypercalcemia. Hypokalemia creates symptoms of cramps, fatigue, myalgias, constipation, and muscular weakness. As potassium levels decrease, action potentials in the muscles decrease, resulting in impaired contractions. Paresthesias (D) are seen in both hypermagnesemia and



hypocalcemia. Increased excitability resulting from hypermagnesemia in muscles results in paresthesia and may be observed in the face. Tetany (E) is a symptom of hypocalcemia, not hypercalcemia. Tetany is described as recurrent muscle spasms. Hypocalcemia causes enhanced excitability of muscles and nerves, which may lead to tetany.